

Hamilton and Phase Space

Trade n second-order equations for $2n$ first-order ones, and the geometry starts doing the work.

WHERE WE ARE

Chapter 1.2 handed you a machine. Feed it one scalar function $L(q, \dot{q}, t)$ and it returns n second-order differential equations, one per degree of freedom, in a form that survives any change of coordinates you care to make. That machine is not going away: it is the formulation that generalises to special relativity (Chapter 2.5), to fields (Chapter 5.2), and to curved spacetime (Chapter 3.6), because the action is a *scalar* and scalars do not care what you call your coordinates.

This chapter builds a second machine out of the same materials. It costs one change of variables — swap the velocities $\dot{q}^i$ for the momenta $p_i = \partial L / \partial \dot{q}^i$ — and it buys three things that the Lagrangian formulation does not give you. First, $2n$ first-order equations instead of n second-order ones, which means the state of the system is a single point and the dynamics is a flow. Second, a theorem — Liouville's — that says that flow is incompressible, which is the foundation of statistical mechanics and, read correctly, the classical ancestor of the statement that quantum evolution is unitary. Third, and this is the one to keep your eye on, an operation on functions called the **Poisson bracket**, with the property that

replacing it by $\frac{1}{i\hbar} [\ , \]$ and changing nothing else turns classical mechanics into quantum mechanics.

That is what Part I is for. Chapter 4.6 will spend one line doing the substitution and the rest of the chapter reaping it, and the reason that line is short is that this chapter is long. Here is the summary you should finish with: **the Lagrangian formulation is the one that generalises; the Hamiltonian formulation is the one that quantizes.** You need both, for different reasons, and mistaking one for the other is a standard way to get stuck.

TOOLS YOU'LL NEED — Chapter 0.4: the determinant as a volume factor, the trace, and $\det(I + \epsilon A) = 1 + \epsilon \operatorname{tr} A + O(\epsilon^2)$ — §5 is that identity. Chapter 0.5: quadratic forms and positive definiteness, which is exactly the condition §1 needs. Chapter 0.6: the Hessian, convexity, the multivariable chain rule, and Clairaut's theorem on mixed partials — §5, §6 and §7 each turn on Clairaut and nothing else. Chapter 0.7: divergence as the trace of the Jacobian, hence as the fractional rate of change of volume carried by a flow. §5 is one substitution into that result. Chapter 0.8: Picard–Lindelöf and the uniqueness of solutions, the harmonic oscillator, the phase-space ellipse, and the pendulum's separatrix. Chapter 1.2: Euler–Lagrange, the canonical momentum $p_i = \partial L / \partial \dot{q}^i$, and Problem 2's Beltrami identity — which, you are about to discover, was the Hamiltonian all along.

1 · The Legendre transform, properly

Chapter 1.2's Problem 2 asked you to show that when L has no explicit time dependence the quantity

$$E = \dot{q} \frac{\partial L}{\partial \dot{q}} - L \quad (1.3.1)$$

is conserved. We are about to promote that combination from a conserved quantity to the central object of a whole formulation of mechanics, and it will be a great deal less mysterious if you first see what kind of operation it is. It is not an ad hoc grouping of terms. It is an instance of a standard construction with a clean geometric meaning, and the same construction is what turns the internal energy of a gas into its free energy. So: §1 is mathematics, and there is no mechanics in it at all.

1.1 · Two ways to describe a curve

Take a smooth, strictly convex function $f : \mathbb{R} \rightarrow \mathbb{R}$ — for now just picture a parabola. There are two ways to say which function it is.

By points. List the pairs $(x, f(x))$. This is the description you grew up with.

By tangent lines. For each slope p , there is exactly one line of that slope tangent to the graph. Give me the family of all those lines — for each p , its intercept — and I can reconstruct the curve, because *a convex curve is the upper envelope of its own tangent lines*. Every tangent line lies below the graph and touches it once; the graph is what you get by taking, at each x , the highest of them.

Both descriptions contain the same information. The **Legendre transform** is the dictionary between them. That is the whole idea, and the formula is going to be an immediate consequence rather than a definition to memorise.

Set it up. A line of slope p has the form $y = px - c$, where $-c$ is its y -intercept; we carry the minus sign because it makes every formula below come out with a plus. Push this line down from above — increase c — until it first touches the graph. Touching means the vertical gap between line and graph has shrunk to zero somewhere; before that, the line sits above the graph everywhere. So the critical value of c is

$$g(p) \equiv \sup_x [px - f(x)]. \quad (1.3.2)$$

Read the bracket as: *how far the line $y = px$ rises above the graph at the point x* . Take the largest such gap. That is the amount you must lower $y = px$ to make it tangent, and it is therefore precisely —(intercept of the tangent line of slope p).

WHAT G IS, IN ONE SENTENCE

$g(p)$ is minus the y -intercept of the tangent line to f whose slope is p . The function g is the "by tangent lines" description of f , packaged so that its argument is the slope.

Everything else in §1 — the formula $p = f'(x)$, the involution $g^{**} = f$, the thermodynamic potentials, and $H = p\dot{q} - L$ — is a consequence of that sentence.

1.2 · Where the supremum sits

Now do the calculus. The bracket in (1.3.2) is, for fixed p , an ordinary function of the single variable x . Chapter 0.6's rule for a stationary point applies:

$$\frac{\partial}{\partial x} [px - f(x)] = p - f'(x) = 0 \quad \implies \quad \boxed{p = f'(x)}. \quad (1.3.3)$$

So the maximising x is the point where the graph's own slope equals p — which is exactly what "tangent line of slope p " meant. Nothing has been assumed yet except differentiability. Two questions remain, and they are what convexity is for.

Is the stationary point a maximum? The second derivative of the bracket is $-f''(x)$, which is negative wherever $f'' > 0$. So yes, provided f is strictly convex.

Is there exactly one? The equation $f'(x) = p$ must have a unique solution for each p in range. Since $f'' > 0$ means f' is strictly increasing, f' is injective, and so it is invertible on its image. Again convexity, and again it is doing genuine work rather than decorating the theorem.

Write $x(p)$ for that unique maximiser. Then

$$g(p) = px(p) - f(x(p)), \quad \text{where } f'(x(p)) = p. \quad (1.3.4)$$

Put (1.3.4) beside (1.3.1) and the shape is already the same: *slope times variable, minus the function*. We will make that identification precise in §2.

1.3 · The transform is its own inverse

Here is the property that makes the construction worth having. Differentiate (1.3.4) with respect to p , using the product rule and the chain rule:

$$\frac{dg}{dp} = x(p) + p \frac{dx}{dp} - f'(x(p)) \frac{dx}{dp} = x(p), \quad (1.3.5)$$

because $f'(x(p)) = p$ makes the last two terms cancel exactly. That cancellation is not a coincidence and it will happen again, in §3, for the same reason: *we are differentiating at a stationary point, so the motion of the stationary point itself contributes nothing at first order*.

Now look at what (1.3.5) says next to (1.3.3):

$$f'(x) = p, \quad g'(p) = x. \quad (1.3.6)$$

Perfectly symmetric. The transform swaps the roles of the variable and the slope. And since $g''(p) = dx/dp = 1/f''(x) > 0$, the transform of a strictly convex function is strictly convex, so we may transform again. Doing so:

$$g^*(x) = \sup_p [xp - g(p)], \quad (1.3.7)$$

whose supremum sits where $g'(p) = x$, i.e. at the p with $x(p) = x$, i.e. at $p = f'(x)$. Evaluating there,

$$g^*(x) = x f'(x) - [f'(x)x - f(x)] = f(x). \quad (1.3.8)$$

The Legendre transform is an **involution** on strictly convex functions: do it twice and you are back where you started. That is what "dictionary" meant — no information is lost going from the point description to the tangent-line description, because you can always go back.

■ One statement here is quoted rather than proved. For a function that is *not* convex, the transform still exists (the supremum in (1.3.2) is well defined whenever it is finite) but the involution fails: the double transform g^* returns the **convex hull** of f — the largest convex function lying below it — and any non-convex wiggles are erased permanently. This is the Fenchel–Moreau theorem, and it is the precise sense in which "the tangent lines determine the curve" needs convexity. We will need it once, in §2.3, to know exactly which Lagrangians the Hamiltonian formulation applies to.

1.4 · A worked scalar example

Take $f(x) = \frac{1}{4}x^4$ on $x \geq 0$, which is strictly convex there ($f'' = 3x^2 > 0$ for $x \neq 0$). Then $p = f'(x) = x^3$, so $x = p^{1/3}$, and

$$g(p) = p \cdot p^{1/3} - \frac{1}{4}(p^{1/3})^4 = p^{4/3} - \frac{1}{4}p^{4/3} = \frac{3}{4}p^{4/3}. \quad (1.3.9)$$

Check the involution without appealing to the theorem. $g'(p) = \frac{3}{4} \cdot \frac{4}{3}p^{1/3} = p^{1/3} = x$ ✓, matching (1.3.6). And

$$g^*(x) = \sup_p \left[xp - \frac{3}{4}p^{4/3} \right] \quad \text{at} \quad p = x^3 \quad \implies \quad x^4 - \frac{3}{4}x^4 = \frac{1}{4}x^4 = f(x). \quad \checkmark \quad (1.3.10)$$

Numerically, at $p = 2$ the supremum of $2x - \frac{1}{4}x^4$ over a fine grid is 1.8898816, and $\frac{3}{4} \cdot 2^{4/3} = 1.8898816$. The formula is measuring something real.

FAMILIAR GROUND

You have used this transform for years under a different name, and knowing that it is the same operation makes the zoo of thermodynamic potentials collapse into one idea.

Start from the internal energy $U(S, V)$ of a simple system, whose differential is the first law,

$$dU = T dS - P dV, \quad \text{so} \quad T = \left. \frac{\partial U}{\partial S} \right|_V, \quad P = - \left. \frac{\partial U}{\partial V} \right|_S.$$

Temperature is the *slope* of U with respect to entropy. Now suppose you would rather label states by T than by S — which is what a laboratory forces on you, since thermometers exist and entropy meters do not. That is exactly the swap (1.3.6) performs. Define

$$F(T, V) = U(S, V) - TS \quad \text{evaluated at the } S \text{ where } \frac{\partial U}{\partial S} = T,$$

and the Helmholtz free energy is -1 times the Legendre transform of U in the variable S . (Physicists carry the extra minus sign so that F is *minimised* at equilibrium; the mathematical content is identical.) Its differential follows in one line:

$$dF = dU - T dS - S dT = -S dT - P dV,$$

where the $T dS$ terms cancel — the same cancellation as (1.3.5), for the same reason. The natural variables of F are (T, V) , and $S = -\partial F / \partial T$ is the inverse relation (1.3.6).

Every other potential is the same move on a different slot. Transform $V \rightarrow P$ instead and you get the enthalpy $\mathcal{H} = U + PV$ with $d\mathcal{H} = T dS + V dP$; transform both and you get the Gibbs free energy $G = \mathcal{H} - TS = U - TS + PV$ with $dG = -S dT + V dP$, which is why G is the one a chemist uses — its natural variables are the two a bench can control. Convexity is not decoration here either: $\partial^2 U / \partial S^2 = \partial T / \partial S = T / C_V > 0$ is thermodynamic stability, and it is exactly the condition (1.3.3) needed for the transform to be invertible. A system with negative heat capacity has no well-defined free energy, which is not a mathematical curiosity — self-gravitating systems have exactly that pathology, and it is why a star heats up when it loses energy.

In §2 we do the identical operation on $L(q, \dot{q})$, swapping $\dot{q}$ for its slope $p = \partial L / \partial \dot{q}$. Same transform, same cancellation, same inverse relation. Only the letters change.

▼ Grind box — Young's inequality, and what the supremum is really enforcing

Definition (1.3.2) says $g(p) \geq px - f(x)$ for every x , with equality only at the maximiser. Rearranged, that is

$$px \leq f(x) + g(p) \quad \text{for all } x, p,$$

with equality precisely when $p = f'(x)$. This is **Young's inequality**, and it is not a separate theorem — it is the definition of the transform, read sideways.

The classic case is $f(x) = x^a/a$ with $a > 1$ on $x \geq 0$. Then $p = x^{a-1}$, so $x = p^{1/(a-1)}$ and

$$g(p) = p^{a/(a-1)} - \frac{1}{a} p^{a/(a-1)} = \frac{a-1}{a} p^{a/(a-1)} = \frac{p^b}{b}, \quad \frac{1}{a} + \frac{1}{b} = 1,$$

since $b \equiv a/(a-1)$ satisfies $1/b = (a-1)/a = 1 - 1/a$. So Young's inequality reads $px \leq x^a/a + p^b/b$ — the inequality from which Hölder's and Minkowski's inequalities are built, and the reason conjugate exponents come in pairs summing to 1 in their reciprocals. Our worked example was $a = 4$, $b = 4/3$. ✓

What breaks without convexity. Take $f(x) = \frac{1}{4}x^4 - \frac{1}{2}x^2$, the double well, which has $f'' = 3x^2 - 1 < 0$ on $|x| < 1/\sqrt{3}$. For each slope p there can now be *three* points with $f'(x) = p$, and (1.3.2) selects whichever gives the largest gap — the global one. The information about the other two is discarded, and the double transform g^* comes back as the convex hull: the double well with its non-convex middle replaced by a straight segment. This is not a defect. It is exactly the *Maxwell construction* of a first-order phase transition: the flat segment is the coexistence region, the tie-line joining liquid and gas, and its slope is the common tangent — the shared value of the chemical potential. Legendre-transforming a non-convex free energy and getting a flat piece is the phase transition. Chapter 6.6 meets the same picture again with the Higgs potential.

The condition in n dimensions. With several variables, $p_i = \partial f / \partial x^i$ is invertible for x precisely when the Jacobian of that map — the **Hessian** $\partial^2 f / \partial x^i \partial x^j$ — is nonsingular, by the inverse function theorem of Chapter 0.6. Strict convexity in the multivariable sense means the Hessian is positive definite (Chapter 0.5), which is more than enough. Hold that thought: in §2 the Hessian in question is $\partial^2 L / \partial \dot{q}^i \partial \dot{q}^j$, which Chapter 1.2's grind box already showed is the positive-definite mass matrix whenever the kinetic energy is an honest quadratic form.

2 · Momentum and the Hamiltonian

2.1 · The canonical momentum is not $m\mathbf{v}$

Chapter 1.2 defined, for any Lagrangian and any coordinates,

$$p_i \equiv \frac{\partial L}{\partial \dot{q}^i}, \quad (1.3.11)$$

the **canonical momentum conjugate to q^i** , and noted that this makes the Euler–Lagrange equation read $\dot{p}_i = \partial L / \partial q^i$. In Cartesian coordinates with $L = \frac{1}{2}m\dot{\mathbf{x}}^2 - V(\mathbf{x})$, (1.3.11) returns $m\dot{x}^i$, which is why the name is not absurd. But that agreement is a special case, and it fails in two different ways that both matter.

It need not have the dimensions of momentum. For the pendulum with coordinate θ , $L = \frac{1}{2}m\ell^2\dot{\theta}^2 + mg\ell \cos \theta$ gives $p_\theta = m\ell^2\dot{\theta}$ — an angular momentum, with units of J s, not kg m s⁻¹. The name "conjugate" is the point: p_i is whatever pairs with q^i so that $p_i dq^i$ has the dimensions of action. Chapter 4.6's uncertainty relation $\Delta q \Delta p \geq \hbar/2$ is dimensionally consistent for every conjugate pair for exactly this reason.

It need not be $m\mathbf{v}$ even in Cartesian coordinates. This is the one worth being careful about. Chapter 2.6 will derive, from relativity, that a particle of charge e in an electromagnetic field described by the scalar potential ϕ and vector potential $\mathbf{A}$ has the Lagrangian

$$L = \frac{1}{2}m \mathbf{v} \cdot \mathbf{v} - e\phi(\mathbf{x}, t) + e \mathbf{v} \cdot \mathbf{A}(\mathbf{x}, t). \quad (1.3.12)$$

Worked Example 1 checks that this reproduces the Lorentz force, so it is the right Lagrangian. Apply (1.3.11) to it, component by component, remembering that $\mathbf{A}$ depends on position but not on velocity:

$$p_i = \frac{\partial}{\partial v^i} \left[\frac{1}{2}m v^j v^j + e v^j A_j \right] = m v^i + e A_i \quad \Longleftrightarrow \quad \mathbf{p} = m\mathbf{v} + e\mathbf{A}. \quad (1.3.13)$$

The canonical momentum and the mechanical momentum $m\mathbf{v}$ differ by $e\mathbf{A}$. This is the first time the vector potential has entered mechanics in this book, and it will not leave: Chapter 2.6 shows $\mathbf{A}$ is the spatial part of a four-vector A^μ and that Maxwell's equations are its equations of motion, and Chapter 6.3 shows that the entire structure of the Standard Model follows from demanding that the substitution $\mathbf{p} \rightarrow \mathbf{p} - e\mathbf{A}$ be forced on you by a symmetry principle. The name for that substitution is **minimal coupling**, and you have just watched it appear, three parts early, as an unremarkable partial derivative.

△ WHY THIS ISN'T OBVIOUS

It is tempting to file (1.3.13) under "notation": surely $m\mathbf{v}$ is the real momentum, and $\mathbf{p}$ is a bookkeeping device. That is exactly backwards, and the difference is measurable.

The gauge problem. Chapter 1.2's Problem 4 showed that $\mathbf{A} \rightarrow \mathbf{A} + \nabla\chi$, $\phi \rightarrow \phi - \partial_t\chi$ changes L by a total time derivative and therefore changes no physics. But it changes $\mathbf{p}$, by $e\nabla\chi$. So the canonical momentum is *gauge-dependent* — it is not, by itself, an observable. Meanwhile $m\mathbf{v} = \mathbf{p} - e\mathbf{A}$ is gauge-invariant, because $\mathbf{A}$ shifts by exactly the same amount. The mechanical momentum is what you measure; the canonical momentum is what the formalism runs on. Both are needed and they are not the same object.

And yet it is $\mathbf{p}$ that gets quantised. In Chapter 4.6 the operator conjugate to position is $\hat{\mathbf{p}} = -i\hbar\nabla$, and the rule for which classical quantity it represents is fixed by the bracket $\{q^i, p_j\} = \delta_j^i$ of §6 — a statement about the *canonical* momentum. Feed (1.3.13) into a Hamiltonian and the kinetic term becomes $(\hat{\mathbf{p}} - e\mathbf{A})^2/2m$, i.e. $-i\hbar\nabla \rightarrow -i\hbar\nabla - e\mathbf{A}$ acting on the wavefunction. Every magnetic effect in quantum mechanics — Landau levels, the Aharonov–Bohm phase, the Zeeman effect, superconducting flux quantisation — comes out of that one replacement, and none of them would if $m\mathbf{v}$ were the thing you quantised.

■ The sharpest experimental statement, quoted forward to Chapter 6.3: in the Aharonov–Bohm effect an electron beam is split around a solenoid and recombined. Outside the solenoid the magnetic field is exactly zero, so a particle described by $m\mathbf{v}$ alone can feel nothing. The interference fringes nonetheless shift, by an amount $e \oint \mathbf{A} \cdot d\ell / \hbar$ — the enclosed flux, in units of $\hbar/e$. The potential is not a computational convenience. It is where the physics lives, and $\mathbf{p} = m\mathbf{v} + e\mathbf{A}$ is the first hint.

2.2 · The Hamiltonian

Now perform §1's transform on L , taking $\dot{q}^i$ as the variable and holding q and t fixed. The conjugate slope is $p_i = \partial L / \partial \dot{q}^i$ by (1.3.11), and the transform (1.3.4) is

$$H(q, p, t) \equiv \sum_i p_i \dot{q}^i - L(q, \dot{q}, t), \quad (1.3.14)$$

where — and this is the entire content of the definition — every $\dot{q}^i$ on the right is to be eliminated in favour of the p_i by inverting (1.3.11). The result is a function of q, p and t only. This object is the **Hamiltonian**, and comparing it with (1.3.1) you can see that Chapter 1.2's Problem 2 was computing it: *the conserved quantity of a time-independent Lagrangian is the Hamiltonian*.

The inversion step is not free, and §1 told us exactly what it needs. Solving $p_i = \partial L / \partial \dot{q}^i$ for $\dot{q}$ requires the Hessian

$$W_{ij} \equiv \frac{\partial^2 L}{\partial \dot{q}^i \partial \dot{q}^j} \quad (1.3.15)$$

to be invertible. Chapter 1.2's grind box on the double pendulum established that when the coordinate change from Cartesians is time-independent, $T = \frac{1}{2} \sum_{ij} M_{ij}(q) \dot{q}^i \dot{q}^j$ with M symmetric positive definite, so $W = M$ and the condition holds automatically. ■ When it fails — when L is linear in some velocity, so that the corresponding row of W vanishes — the transform is not invertible and the system is called **constrained**. Dirac built a whole formalism for that case; it is quoted here and not developed, but you should know it is where gauge theories live (Chapter 6.3: the Lagrangian of electromagnetism has no $\dot{A}_0$ at all) and where the reparametrisation-invariant string action lives (Chapter 7.3). Every fundamental theory in Part VI and VII is a constrained system, so the honest statement is: the clean Legendre transform of this section is the easy case, and it is the one that teaches you what the hard case is deforming.

2.3 · When $H = T + V$, and when it isn't

You have surely been told that the Hamiltonian is the total energy. It often is. It is not a definition, it is a theorem, and the theorem has two hypotheses that fail in ordinary situations.

THEOREM

Suppose (i) V does not depend on the velocities, and (ii) T is a *homogeneous quadratic form* in the generalized velocities, $T = \frac{1}{2} \sum_{ij} M_{ij}(q) \dot{q}^i \dot{q}^j$. Then $H = T + V$.

Proof. With $L = T - V$ and hypothesis (i), $p_i = \partial L / \partial \dot{q}^i = \partial T / \partial \dot{q}^i$. Hypothesis (ii) says T is homogeneous of degree 2: replacing every $\dot{q}$ by $\lambda \dot{q}$ multiplies T by λ^2 . Differentiate that identity, $T(q, \lambda \dot{q}) = \lambda^2 T(q, \dot{q})$, with respect to λ and set $\lambda = 1$; the chain rule gives

$$\sum_i \dot{q}^i \frac{\partial T}{\partial \dot{q}^i} = 2T \quad (1.3.16)$$

(this is Euler's theorem on homogeneous functions, derived rather than quoted — the grind box does the general degree). Therefore

$$H = \sum_i p_i \dot{q}^i - L = \sum_i \dot{q}^i \frac{\partial T}{\partial \dot{q}^i} - (T - V) = 2T - T + V = T + V. \quad (1.3.17)$$

■ Now break the hypotheses, using a system Chapter 1.2 already built. Problem 3 there put a bead on a hoop of radius R spun about its vertical diameter at a rate Ω fixed by a motor, with

$$L = \frac{1}{2} m R^2 \dot{\theta}^2 + \frac{1}{2} m R^2 \Omega^2 \sin^2 \theta + m g R \cos \theta. \quad (1.3.18)$$

The middle term is part of the kinetic energy — it is the speed the bead has purely because the hoop is carrying it around — but it contains *no* $\dot{\theta}$. So T is not homogeneous of degree 2 in the velocity; hypothesis (ii) fails. Compute honestly: $p_\theta = m R^2 \dot{\theta}$, so $\dot{\theta} = p_\theta / m R^2$, and

$$\begin{aligned}
H = p_\theta \dot{\theta} - L &= \frac{p_\theta^2}{2mR^2} - \frac{1}{2}mR^2\Omega^2 \sin^2 \theta - mgR \cos \theta, \\
T + V &= \frac{p_\theta^2}{2mR^2} + \frac{1}{2}mR^2\Omega^2 \sin^2 \theta - mgR \cos \theta.
\end{aligned}
\tag{1.3.19}$$

They differ by $mR^2\Omega^2 \sin^2 \theta$, which changes as the bead slides. H is conserved (it has no explicit t); $T + V$ is not, because the motor is doing work to hold Ω fixed. **Conserved and "the energy" are different predicates, and H is the first one.** Notice also what H is here: kinetic energy of the θ motion plus the effective potential V_{eff} that Chapter 1.2 found — the centrifugal term has flipped sign on its way through the transform and become part of the potential. That is not sleight of hand; it is (1.3.16) refusing to apply to a degree-zero term.

▼ Grind box — Euler's theorem in general, the three-piece decomposition, and the relativistic particle

Euler's theorem. Let F be homogeneous of degree k in the variables u^i , meaning $F(\lambda u) = \lambda^k F(u)$ for all $\lambda > 0$. Differentiate both sides with respect to λ using the chain rule:

$$\sum_i u^i \frac{\partial F}{\partial u^i} \Big|_{\lambda u} = k \lambda^{k-1} F(u),$$

and set $\lambda = 1$ to get $\sum_i u^i \partial F / \partial u^i = kF$. That is the whole proof. Degree 2 gives (1.3.16); degree 1 gives $\sum u^i \partial_i F = F$; degree 0 gives $\sum u^i \partial_i F = 0$.

The general Lagrangian. Split $L = L_2 + L_1 + L_0$ by degree in $\dot{q}$. Then $p_i \dot{q}^i = 2L_2 + L_1$ by the three cases above, so

$$H = (2L_2 + L_1) - (L_2 + L_1 + L_0) = L_2 - L_0.$$

Every degree-one piece drops out of H entirely. Three consequences, all checkable against what we have already done:

- Ordinary mechanics: $L_2 = T$, $L_1 = 0$, $L_0 = -V$, giving $H = T + V$. ✓
- Rotating hoop: $L_2 = \frac{1}{2}mR^2\dot{\theta}^2$, $L_1 = 0$, $L_0 = \frac{1}{2}mR^2\Omega^2 \sin^2 \theta + mgR \cos \theta$, giving exactly (1.3.19). ✓ The centrifugal term's sign flip is the minus in $L_2 - L_0$.
- Charged particle: $L_2 = \frac{1}{2}mv^2$, $L_1 = e \mathbf{v} \cdot \mathbf{A}$, $L_0 = -e\phi$, giving $H = \frac{1}{2}mv^2 + e\phi$. The magnetic term vanishes from H — which is the statement that **magnetic forces do no work**, derived rather than asserted. It reappears the moment you express H in terms of $\mathbf{p}$ rather than $\mathbf{v}$, because $\frac{1}{2}mv^2 = (\mathbf{p} - e\mathbf{A})^2/2m$. Worked Example 1 does this properly.

The relativistic free particle. Chapter 1.2's table listed $S = -mc^2 \int d\tau$, i.e. in one dimension $L = -mc^2 \sqrt{1 - \dot{x}^2/c^2}$, which is not of the form $T - V$ and is not homogeneous of any degree. Turn the crank anyway:

$$p = \frac{\partial L}{\partial \dot{x}} = -mc^2 \cdot \frac{-\dot{x}/c^2}{\sqrt{1 - \dot{x}^2/c^2}} = \frac{m\dot{x}}{\sqrt{1 - \dot{x}^2/c^2}} = \gamma m\dot{x}.$$

Then, using $1/\gamma = \sqrt{1 - \dot{x}^2/c^2}$,

$$H = p\dot{x} - L = \gamma m \dot{x}^2 + \frac{mc^2}{\gamma} = \frac{m(\gamma^2 \dot{x}^2 + c^2)}{\gamma} = \frac{mc^2 \gamma^2}{\gamma} = \gamma mc^2,$$

where the middle step used $\gamma^2 \dot{x}^2 + c^2 = c^2 \gamma^2 (\dot{x}^2/c^2 + 1/\gamma^2) = c^2 \gamma^2$. So $H = \gamma mc^2$: the relativistic energy, including the rest energy, produced by a Legendre transform and nothing else. Expressed in the right variable — which is what H demands — we use $p^2 c^2 = \gamma^2 m^2 \dot{x}^2 c^2$ and the same identity to get

$$H(p) = \sqrt{p^2 c^2 + m^2 c^4}.$$

That is the relation Chapter 2.5 derives from four-vectors and Chapter 5.4 quantises into the Dirac equation. It fell out of §1's transform applied to a square root. Note in passing that $\partial^2 L / \partial \dot{x}^2 = m\gamma^3 > 0$, so the Hessian condition (1.3.15) holds and the transform is legitimate — the same positivity Chapter 1.2's second-variation grind box used to show that a relativistic worldline maximises proper time.

3 · Hamilton's equations, derived

We have a new function $H(q, p, t)$. We do not yet have equations of motion in terms of it. There are two ways to get them, and they are worth seeing both because they emphasise different things.

3.1 · First route: take the differential

Chapter 0.6's total differential is the statement that for any smooth function of several variables, the first-order change is the sum of the partial derivatives times the changes in their arguments. Apply it to the *right-hand side* of (1.3.14), treating q , $\dot{q}$ and t as the independent variables there:

$$dH = \sum_i \left(\dot{q}^i dp_i + p_i d\dot{q}^i \right) - \sum_i \left(\frac{\partial L}{\partial q^i} dq^i + \frac{\partial L}{\partial \dot{q}^i} d\dot{q}^i \right) - \frac{\partial L}{\partial t} dt. \quad (1.3.20)$$

Now look at the two terms containing $d\dot{q}^i$. Their coefficients are p_i and $-\partial L / \partial \dot{q}^i$, and by the definition (1.3.11) those are equal and opposite. **They cancel identically.**

That cancellation is the entire trick, and it deserves a sentence of its own: *the Legendre transform is engineered so that the variable you are trying to get rid of drops out of the differential*. It is the same cancellation as (1.3.5) and the same one as $dF = dU - TdS - SdT$ in the thermodynamics callout. Not an analogy — one theorem, appearing for the third time.

What survives is

$$dH = \sum_i \dot{q}^i dp_i - \sum_i \frac{\partial L}{\partial q^i} dq^i - \frac{\partial L}{\partial t} dt, \quad (1.3.21)$$

which contains only dq , dp and dt — confirming that H really is a function of (q, p, t) and not secretly of the velocities. But if H is a function of (q, p, t) then its differential is also, by definition,

$$dH = \sum_i \frac{\partial H}{\partial q^i} dq^i + \sum_i \frac{\partial H}{\partial p_i} dp_i + \frac{\partial H}{\partial t} dt. \quad (1.3.22)$$

Two expressions for the same differential, in terms of the same independent increments. Their coefficients must match one by one. Comparing dp_i , then dq^i , then dt :

$$\frac{\partial H}{\partial p_i} = \dot{q}^i, \quad \frac{\partial H}{\partial q^i} = -\frac{\partial L}{\partial q^i}, \quad \frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t}. \quad (1.3.23)$$

Only the middle one still mentions L . Kill it with the Euler–Lagrange equation, which says $\partial L / \partial q^i = \frac{d}{dt}(\partial L / \partial \dot{q}^i) = \dot{p}_i$. And there they are:

$$\boxed{\dot{q}^i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q^i}, \quad \frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t}.} \quad (1.3.24)$$

Hamilton's equations. Count what happened: n second-order equations became $2n$ first-order ones. Nothing was gained or lost in information — a second-order equation always splits into two first-order ones, and the standard way to do that is to call $\dot{q}$ a new variable. What is special here is *which* new variable was chosen. Choosing $p = \partial L / \partial \dot{q}$ rather than $\dot{q}$ itself is what produces the near-symmetry of (1.3.24), and that symmetry is worth everything that follows.

One immediate dividend, free: if H has no explicit time dependence then

$$\frac{dH}{dt} = \sum_i \left(\frac{\partial H}{\partial q^i} \dot{q}^i + \frac{\partial H}{\partial p_i} \dot{p}_i \right) + \frac{\partial H}{\partial t} = \sum_i \left(-\dot{p}_i \dot{q}^i + \dot{q}^i \dot{p}_i \right) + 0 = 0. \quad (1.3.25)$$

H is conserved, by cancellation between the two halves of (1.3.24). That is Chapter 1.2's Beltrami identity again, now in three lines instead of five, and the minus sign that made the two equations *not* symmetric is precisely what made the cancellation happen. Remember that; §5 and §6 are both that minus sign.

3.2 · Second route: vary the action in phase space

The first derivation used the Euler–Lagrange equation as an input. Here is a derivation that does not, and that treats q and p on an equal footing from the start.

Rearrange (1.3.14) as $L = \sum_i p_i \dot{q}^i - H$ and substitute it into the action. Then *declare* $q^i(t)$ and $p_i(t)$ to be $2n$ independent functions, with no relation between them assumed:

$$S[q, p] = \int_{t_1}^{t_2} \left[\sum_i p_i \dot{q}^i - H(q, p, t) \right] dt. \quad (1.3.26)$$

This is a functional on a space of paths in the $2n$ -dimensional space of (q, p) , and we ask for its stationary points exactly as Chapter 1.2 did. Vary both: $q \rightarrow q + \epsilon \eta$, $p \rightarrow p + \epsilon \xi$. Note the asymmetry that is about to matter: $\dot{q}$ appears in (1.3.26) but $\dot{p}$ does not.

$$\delta S = \int_{t_1}^{t_2} \sum_i \left[\xi_i \dot{q}^i + p_i \dot{\eta}^i - \frac{\partial H}{\partial q^i} \eta^i - \frac{\partial H}{\partial p_i} \xi_i \right] dt. \quad (1.3.27)$$

Only one term carries a derivative of a variation, and Chapter 1.2's move three handles it: integrate $\int p_i \dot{\eta}^i dt$ by parts to get $[p_i \eta^i]_{t_1}^{t_2} - \int \dot{p}_i \eta^i dt$. Collecting the coefficients of η and ξ separately,

$$\delta S = \left[\sum_i p_i \eta^i \right]_{t_1}^{t_2} + \int_{t_1}^{t_2} \sum_i \left[\left(\dot{q}^i - \frac{\partial H}{\partial p_i} \right) \xi_i - \left(\dot{p}_i + \frac{\partial H}{\partial q^i} \right) \eta^i \right] dt. \quad (1.3.28)$$

Kill the boundary term by requiring $\eta^i(t_1) = \eta^i(t_2) = 0$ — fix the positions at both ends, as before. Note carefully that **no condition on ξ is needed**: p never appears differentiated, so it generates no boundary term and is free at the endpoints. That is not a technicality. It says the natural boundary-value problem in phase space is "specify where you start and where you finish", not "specify the whole initial state" — and Chapter 5.8's path integral, which sums over paths with fixed endpoints, inherits exactly this asymmetry.

Now apply the fundamental lemma (Chapter 1.2 §3.4) twice: once with ξ arbitrary and $\eta = 0$, once the other way round. Each bracket must vanish separately, giving

$$\dot{q}^i - \frac{\partial H}{\partial p_i} = 0, \quad \dot{p}_i + \frac{\partial H}{\partial q^i} = 0, \quad (1.3.29)$$

which is (1.3.24) again. The two derivations agree, as they must, and each tells you something the other does not: route one tells you the equations are the Legendre transform of Euler–Lagrange, and route two tells you they are themselves a variational principle — one in which position and momentum are independent variables of equal status.

3.3 · The minus sign, and what it is the seed of

Stare at (1.3.24) until the near-symmetry bothers you. The equations for q and p are the same equation, with $q \leftrightarrow p$ — except for one minus sign. That asymmetry is irreducible; you cannot rescale it away, because rescaling $p \rightarrow -p$ just moves the sign to the other equation.

It is convenient to make it explicit. Stack the coordinates into one list of $2n$ numbers,

$$z = (q^1, \dots, q^n, p_1, \dots, p_n), \quad (1.3.30)$$

and define the $2n \times 2n$ matrix

$$\Omega = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}, \quad \Omega^\top = -\Omega, \quad \Omega^2 = -I_{2n}, \quad \det \Omega = 1. \quad (1.3.31)$$

Then both halves of (1.3.24) collapse into a single line:

$$\dot{z} = \Omega \nabla H, \quad (\nabla H)_a = \frac{\partial H}{\partial z^a}. \quad (1.3.32)$$

Check it: the top block of $\Omega \nabla H$ is $+\partial H/\partial p$ and the bottom block is $-\partial H/\partial q$. ✓ The gradient of H points uphill, at right angles to the level sets; Ω rotates that direction by a quarter turn; and the system moves *along* the level sets. Which is why H is conserved — the flow is everywhere perpendicular to the gradient, by construction.

Ω is called the **symplectic form**, and the antisymmetry $\Omega^\top = -\Omega$ is the minus sign, isolated. Compare it with the object that runs Chapter 3.3, the metric $g_{\mu\nu}$, which is *symmetric* and measures lengths. A symplectic form measures oriented areas instead, and it is the reason phase space has a geometry at all. Every structural fact in the rest of this chapter — Liouville's theorem, the Poisson bracket, the $[\hat{q}, \hat{p}] = i\hbar$ of quantum mechanics — is a consequence of the antisymmetry of Ω . So is the fact that phase space is always even-dimensional: an antisymmetric matrix in odd dimensions has $\det \Omega = \det \Omega^\top = \det(-\Omega) = (-1)^{2n+1} \det \Omega$, forcing $\det \Omega = 0$, so no invertible Ω exists there.

4 · Phase space

The space whose points are the $2n$ numbers $z = (q, p)$ is **phase space**. Chapter 0.8 introduced it for the oscillator and promised it would come back; this is where it becomes the setting rather than a picture.

4.1 · One point fixes everything, and trajectories cannot cross

Hamilton's equations (1.3.32) are first order. That is the structural difference from the Lagrangian formulation and it changes what a "state" is. Chapter 0.8's Picard–Lindelöf theorem says that a first-order system $\dot{z} = F(z)$ with F Lipschitz has exactly one solution through each initial point. So:

THE DEFINING PROPERTY OF PHASE SPACE

A single point of phase space determines the entire past and future of the system. Not a point plus a velocity — a point.

Configuration space does not have this property. A pendulum passing through $\theta = 0$ might be moving left or right, fast or slow; " $\theta = 0$ " is not a state. Phase space is precisely the space of things that *are* states, which is why Chapter 0.8 said the list the existence theorem asks for is the list phase space is made of.

There is an immediate geometric consequence, and it is what makes phase portraits legible.

Claim. If H has no explicit time dependence, two distinct trajectories can never cross.

Proof. Suppose the trajectory z_1 passes through the point z_* at time t_1 and the trajectory z_2 passes through the same z_* at time t_2 . Because H does not depend on t , the right-hand side of (1.3.32) depends only on the point, so shifting a solution in time gives another solution: define $\tilde{z}_2(t) = z_2(t + t_2 - t_1)$, which also solves the equations and satisfies $\tilde{z}_2(t_1) = z_*$. Now z_1 and $\tilde{z}_2$ are two solutions with the same value at t_1 , so by uniqueness they are the same solution. Hence z_2 is z_1 retimed: the same curve. ■

So through every point of phase space there passes exactly one curve, and phase space is filled by these curves like the streamlines of a steady fluid — no branching, no merging, no intersections. The only points where curves appear to meet are **fixed points**, where $\nabla H = 0$ and the "trajectory" is a single stationary point. We will meet one shortly.

4.2 · The harmonic oscillator, again

Take $H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2$, which is $T + V$ (both hypotheses of §2.3 hold). Hamilton's equations are

$$\dot{q} = \frac{\partial H}{\partial p} = \frac{p}{m}, \quad \dot{p} = -\frac{\partial H}{\partial q} = -m\omega^2 q. \quad (1.3.33)$$

The first says $p = m\dot{q}$, recovering the mechanical momentum; substituting it into the second gives $m\ddot{q} = -m\omega^2 q$, the oscillator equation. Nothing new — but now read the pair as a flow instead of as a route back to a second-order equation.

The level sets of H are the ellipses of Chapter 0.8 §4.4, and by §3.3 the motion runs along them. To see what the motion actually is, rescale so that the ellipses become circles. Put

$$Q = \sqrt{m\omega} q, \quad P = \frac{p}{\sqrt{m\omega}}, \quad (1.3.34)$$

which has the virtue of preserving areas ($dQ dP = dq dp$) — a property §8 will name. In these variables $H = \frac{\omega}{2}(P^2 + Q^2)$ and (1.3.33) becomes

$$\dot{Q} = \omega P, \quad \dot{P} = -\omega Q. \quad (1.3.35)$$

That is a **rigid rotation** of the whole plane at angular velocity ω , clockwise. Every point, whatever its energy, goes round at the same rate — which is the phase-space statement of the fact that a harmonic oscillator's period is independent of amplitude. Chapter 0.1 observed that i is a rotation by 90° ; here the rotation is generated by Ω , which squares to -1 for the same reason. The oscillator flow is multiplication by $e^{-i\omega t}$, drawn.

4.3 · The pendulum, and the separatrix

Now a system where the flow is not uniform. Take the pendulum of Chapter 1.2 §6.2, with $q = \theta$, $p = m\ell^2\dot{\theta}$, and (using §2.3's theorem, whose hypotheses hold)

$$H = \frac{p^2}{2m\ell^2} + mg\ell(1 - \cos q), \quad (1.3.36)$$

where the constant has been chosen so that $H = 0$ at the bottom. Hamilton's equations are $\dot{q} = p/m\ell^2$ and $\dot{p} = -mg\ell \sin q$, reproducing $\ddot{q} = -(g/\ell) \sin q$. Write $\omega_0^2 = g/\ell$.

Since H is conserved, trajectories are its level sets, and there are three kinds.

Libration. For $0 < H < 2mg\ell$ the bob cannot reach the top: setting $p = 0$ in (1.3.36) gives a turning point at $\cos q = 1 - H/mg\ell$, which has a solution only when $H \leq 2mg\ell$. The level set is a closed curve encircling the origin, and the motion is back-and-forth swinging. Near the origin, expanding $1 - \cos q \approx q^2/2$ recovers the oscillator of §4.2 and the level curves are Chapter 0.8's ellipses.

Rotation. For $H > 2mg\ell$ there is no turning point: p never vanishes, the bob goes over the top and keeps going. The level set is an open curve running across the whole range of q , and since q and $q + 2\pi$ are the same physical configuration, phase space here is really a cylinder and the curve is a loop around it.

The separatrix. Exactly at

$$H_{\text{sep}} = 2mg\ell, \quad (1.3.37)$$

the bob arrives at the inverted position $q = \pi$ with exactly zero momentum. Solving (1.3.36) for p at this energy and using $1 - \cos q = 2 \sin^2(q/2)$,

$$\frac{p^2}{2m\ell^2} = mg\ell(1 + \cos q) = 2mg\ell \cos^2 \frac{q}{2} \implies p = \pm 2m\ell^2 \omega_0 \cos \frac{q}{2}, \quad (1.3.38)$$

the two arcs Chapter 0.8 §8.2 found. They meet at $q = \pm\pi, p = 0$, and that meeting point is a **fixed point** — $\partial H/\partial q = mg\ell \sin \pi = 0$ and $\partial H/\partial p = 0$ — so §4.1's no-crossing theorem is not violated. Linearise about it by writing $q = \pi + u$, so $\sin q = -\sin u \approx -u$:

$$\dot{u} = \frac{p}{m\ell^2}, \quad \dot{p} = +mg\ell u \implies \ddot{u} = +\omega_0^2 u, \quad (1.3.39)$$

whose solutions are $e^{\pm\omega_0 t}$ — one growing, one decaying. This is a **hyperbolic fixed point**, or saddle: the level curves near it are hyperbolas, and the separatrix is the pair of straight lines they asymptote to. The decaying solution is the trajectory that creeps toward the inverted position and takes infinite time to arrive; the growing one is the same trajectory run backwards. Compare the origin, where the same calculation gives $\ddot{u} = -\omega_0^2 u$ and the level curves are closed: an **elliptic fixed point**, or centre. Two fixed points, two signs, two completely different local pictures — and the entire qualitative structure of the pendulum is the statement that they are joined by the separatrix.

4.4 · What the enclosed area means

For a closed orbit — libration — the natural number to attach to it is the area it encloses:

$$\mathcal{A}(E) = \oint_{H=E} p \, dq, \quad (1.3.40)$$

where the line integral around the closed curve equals the enclosed area by Green's theorem (Chapter 0.7 §5.1) — that is what a line integral of $p \, dq$ around a loop is. Two facts make this the right quantity to care about.

First: its derivative is the period. Solve $H(q, p) = E$ for p on the upper branch and differentiate (1.3.40) under the integral sign. Since $\partial p / \partial E$ at fixed q is the reciprocal of $\partial H / \partial p$ at fixed q , and $\partial H / \partial p = \dot{q}$,

$$\frac{d\mathcal{A}}{dE} = \oint \frac{\partial p}{\partial E} dq = \oint \frac{dq}{\partial H / \partial p} = \oint \frac{dq}{\dot{q}} = \oint dt = T(E). \quad (1.3.41)$$

The area is the antiderivative of the period. For the oscillator, Chapter 0.8 computed $\mathcal{A} = 2\pi E / \omega$, so $d\mathcal{A} / dE = 2\pi / \omega = T$ ✓ — and since T does not depend on E there, the area is exactly linear in the energy. For the pendulum it is not: numerical integration of (1.3.40) at $\omega_0 = 1, m\ell^2 = 1$ gives $d\mathcal{A} / dE = 6.449765$ at $E = 0.2$ and 11.633349 at $E = 1.9$, against periods $4K(E/2) / \omega_0$ of 6.449765 and 11.633349 from Chapter 0.8's elliptic-integral formula. Agreement to ten digits, which is the sort of thing worth checking once.

Second: it is what quantum mechanics rations. ▮ Quoted forward to Chapter 4.5. The **Bohr–Sommerfeld condition** states that in the semiclassical regime the allowed orbits are those with

$$\oint p \, dq = \left(n + \frac{1}{2}\right) h, \quad n = 0, 1, 2, \dots \quad (1.3.42)$$

with $h = 2\pi\hbar$ Planck's constant. For the oscillator this reads $2\pi E_n / \omega = (n + \frac{1}{2})h$, i.e. $E_n = (n + \frac{1}{2})\hbar\omega$ — which Chapter 4.5 will derive exactly, with ladder operators and no semiclassical approximation, and get precisely this answer. So the classical phase-space ellipse is not deleted by quantum mechanics; it is quantised. The continuum of orbits becomes a discrete stack of them, one per unit of h of enclosed area, and the ground state is the one with half a unit. **Phase-space area is measured in units of Planck's constant.** That sentence is the reason h has the units it does — J s, the units of $p \, dq$ — and it is the single most useful thing to remember about phase space.

▼ Grind box — the area–period relation done carefully, and the adiabatic invariant

The derivation, without hand-waving. Take a one-dimensional system with $H(q, p)$ and a closed orbit at energy E . On the upper half of the orbit write $p = p_+(q, E)$; the enclosed area is $\mathcal{A}(E) = \oint p \, dq$, traversed so the integral is positive. Differentiating with respect to E moves the curve; the endpoints (the turning points) move too, but $p = 0$ there, so the boundary contribution vanishes and we may differentiate under the integral:

$$\frac{d\mathcal{A}}{dE} = \oint \left(\frac{\partial p}{\partial E} \right)_q dq.$$

Now, differentiating the identity $H(q, p(q, E)) = E$ with respect to E at fixed q gives $(\partial H / \partial p)(\partial p / \partial E) = 1$, so $\partial p / \partial E = 1 / \dot{q}$. Hence the integrand is $dq / \dot{q} = dt$, and integrating once round the orbit gives one period. ■

The action variable. Define $J = \mathcal{A} / 2\pi$. Then $dE / dJ = 2\pi / T = \omega(E)$, the angular frequency. Problem 3 constructs a change of variables in which J is literally the new momentum and the new

coordinate is an angle advancing uniformly at rate ω ; these are **action-angle variables**, and they turn any one-dimensional bound motion into the trivial system $\dot{\vartheta} = \omega, \dot{J} = 0$.

■ One quoted fact, because it explains why (1.3.42) was ever a sensible guess. If a parameter of the system (a pendulum's length, say) is changed *slowly* compared with the period, then E and T both change but J does not, to all orders in the slowness. J is an **adiabatic invariant**. Ehrenfest's argument was that only an adiabatic invariant can sensibly be quantised — if you could change J by gently stretching a wire you could move a system off its allowed levels without a transition — and that is how Bohr and Sommerfeld chose which quantity to set equal to $n\hbar$. The modern statement is Chapter 4.5's adiabatic theorem: a quantum system stays in the n -th eigenstate under slow change, which is the same sentence with J replaced by n .

Numbers. For the pendulum at $m\ell^2 = 1, \omega_0 = 1$, quadrature of (1.3.40) against the elliptic-integral period gives

E	dA/dE (NUMERICAL)	$T = 4K(E/2)$	RELATIVE DIFFERENCE
0.2	6.449765	6.449765	2×10^{-11}
0.8	7.110077	7.110077	1×10^{-12}
1.4	8.301453	8.301453	4×10^{-11}
1.9	11.633349	11.633349	3×10^{-10}

The period diverges as $E \rightarrow 2$ — the separatrix takes infinite time — and (1.3.41) says the area therefore has infinite slope there, while remaining finite. Both statements are visible in the interactive below: near the separatrix, neighbouring orbits have wildly different periods, so a blob of initial conditions is sheared apart. That shearing is the whole show.

5 · Liouville's theorem

This is the centrepiece, and it costs almost nothing because Chapter 0.7 already did the work.

Forget for a moment that (1.3.24) describes mechanics and read it as what it literally is: a formula assigning a velocity vector to every point of a $2n$ -dimensional space. That is a **vector field** in the sense of Chapter 0.7 §1, namely

$$\mathbf{v}(z) = (\dot{q}^1, \dots, \dot{q}^n, \dot{p}_1, \dots, \dot{p}_n) = \left(\frac{\partial H}{\partial p_1}, \dots, \frac{\partial H}{\partial p_n}, -\frac{\partial H}{\partial q^1}, \dots, -\frac{\partial H}{\partial q^n} \right). \quad (1.3.43)$$

Chapter 0.7 spent a section establishing what the divergence of such a field means, and the answer was not "something to do with flux" but something much more useful: *the divergence is the trace of the Jacobian, hence the fractional rate at which a blob carried by the flow changes its volume*,

$$\frac{1}{V} \frac{dV}{dt} = \text{tr } J = \nabla \cdot \mathbf{v}. \quad (1.3.44)$$

So compute the divergence of (1.3.43). It is the sum of $\partial(\text{component})/\partial(\text{corresponding coordinate})$ over all $2n$ coordinates:

$$\begin{aligned} \nabla \cdot \mathbf{v} &= \sum_i \left[\frac{\partial}{\partial q^i} \left(\frac{\partial H}{\partial p_i} \right) + \frac{\partial}{\partial p_i} \left(-\frac{\partial H}{\partial q^i} \right) \right] \\ &= \sum_i \left[\frac{\partial^2 H}{\partial q^i \partial p_i} - \frac{\partial^2 H}{\partial p_i \partial q^i} \right] = 0 \end{aligned} \quad (1.3.45)$$

by the equality of mixed partial derivatives — Clairaut's theorem, Chapter 0.6 §6.1. Every term cancels against its partner, and the cancellation happens because of the minus sign of §3.3 and nothing else. Substituting into (1.3.44):

$$\boxed{\frac{dV}{dt} = 0.} \quad (1.3.46)$$

Liouville's theorem: Hamiltonian flow preserves phase-space volume exactly. Not approximately, not on average, not for special systems. For every Hamiltonian, including time-dependent ones (nothing in the calculation used $\partial H/\partial t = 0$), including chaotic ones, including the entire universe if you are willing to write down its Hamiltonian.

WHAT HAS ACTUALLY BEEN PROVED

Take any region R_0 of phase space — a set of possible initial conditions. Let every point of it evolve for a time t , and call the resulting set R_t . Then $\text{vol}(R_t) = \text{vol}(R_0)$.

The shape of R_t is entirely unconstrained. It may be stretched, sheared, folded, wound into a spiral of a million turns, or drawn out into a filament thinner than any resolution you can afford. The theorem says nothing whatever about shape. It says the measure is exactly invariant, and it is that combination — arbitrary distortion, exactly conserved measure — that makes the theorem worth having.

Three payoffs, in increasing order of importance.

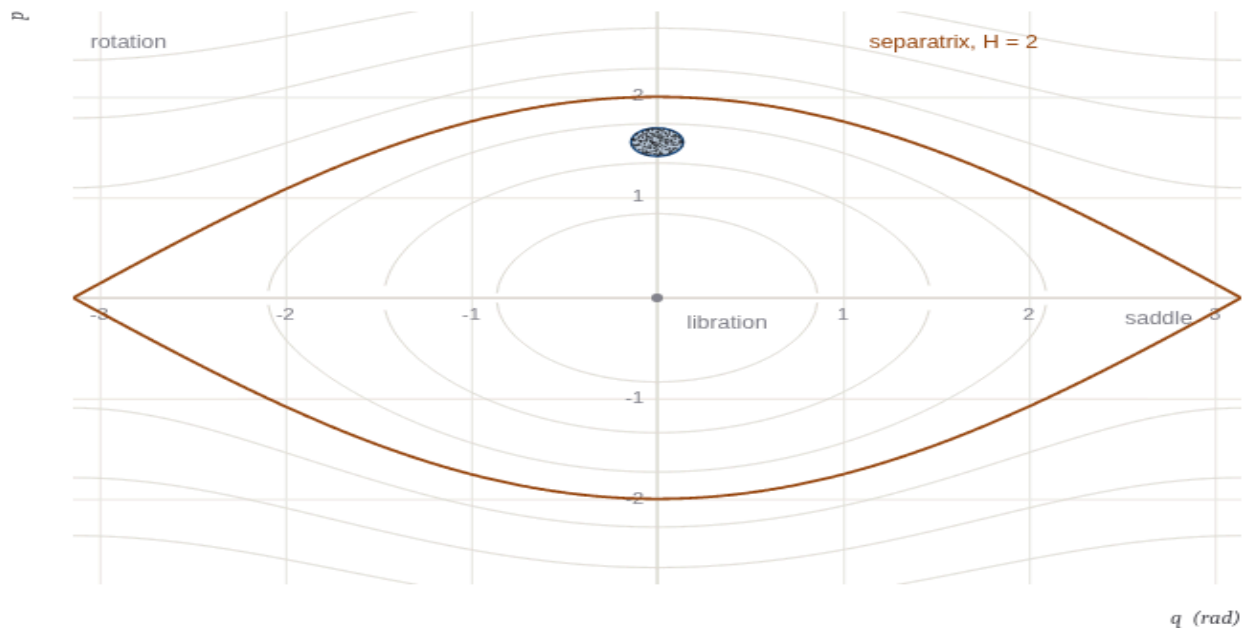
Statistical mechanics has a measure. To do statistical mechanics you must say what "equally likely" means for a continuous system, and that requires a measure on phase space that the dynamics does not distort. Liouville supplies exactly one: $d^n q d^n p$. The microcanonical ensemble — uniform on a surface of constant energy — is well defined

because the flow is incompressible; had the flow contracted volumes, an initially uniform ensemble would pile up and "uniform" would not be a stable notion. Every ensemble average you have ever seen rests on (1.3.46).

There is no classical attractor. A damped oscillator spirals into the origin, so a blob of initial conditions shrinks to a point and volume is destroyed. Liouville therefore says damping is never fundamental: it is what a Hamiltonian system looks like when you have thrown away the degrees of freedom (the air, the wire, the thermal bath) that the energy leaked into. The full system, including the bath, conserves volume. This is the reason a fundamental theory can never contain a friction term, and why every dissipative equation in physics is an effective description.

Information is not destroyed, only stirred. This is the one to keep. Suppose you know the state to within a small blob of volume V_0 — your measurement precision. After evolution the blob has some horrible filamentary shape, but its volume is still V_0 . In principle, running the equations backwards recovers the initial blob exactly: *Hamiltonian evolution is reversible and loses nothing*. Two forward pointers, both of which are this sentence in different clothes.

- Chapter 4.4: quantum time evolution is generated by $e^{-i\hat{H}t/\hbar}$, which is a **unitary** operator — it preserves inner products, hence probabilities, hence distinguishability. Two distinct initial states stay distinct forever. That is the quantum statement of exactly this theorem, with "volume in phase space" replaced by "overlap of state vectors" and the symplectic form Ω replaced by the factor of i .
- Chapters 3.8 and 7.8: the **black-hole information problem** is what happens when the two statements appear to disagree. Hawking's calculation says a black hole evaporates into radiation that depends only on its mass, charge and spin — so the volume of states that could have formed it collapses to a point, and information is destroyed. Something in that argument must be wrong, because unitarity is not a convenience you can trade away, and the search for what is wrong produced holography. The problem is a problem precisely because (1.3.46) and its quantum successor are load-bearing.



play

reset

blob start p_0 $p_0 = 1.55$ (H at blob top = 1.428)

$t = 0.00$ boundary vertices = 400

area now = 0.0615727 initial = 0.0615727

drift = +0.0 ppm – exactly zero is the theorem

Liouville, made visible. The pendulum (1.3.36) in units $m\ell^2 = 1$, $\omega_0 = 1$, so the separatrix (orange) sits at $H = 2$ and the grey curves are other level sets of H . A disc of radius 0.14 — 240 interior points, drawn, plus a 400-vertex boundary polygon — is released centred on the p -axis and every point is advanced by fourth-order Runge–Kutta with step 0.01. **What to do.** Press *play* with the slider near the bottom of its range: the blob is deep inside the libration region where the flow is nearly the rigid rotation of §4.2, and it goes round almost undeformed. Now slide it up toward 1.82, where the top of the blob has energy 1.921 against a separatrix at 2. Different parts of the blob are now on orbits with substantially different periods — (1.3.41) says the period diverges at the separatrix — so the blob is sheared. Within one circuit it is a crescent; within four it is a filament wrapped twice around the level curves and threaded along the separatrix; the dashed grey circle shows where it started. **The number that refuses to move.** The readout is the shoelace area of the boundary polygon, which is exact for a polygon, so it is a measurement and not a formula. The boundary is adaptively refined — points are inserted whenever neighbours drift more than 0.03 apart, and the count is reported — because a fixed polygon would eventually cut corners off a curve that is growing without bound in length. Watch the area sit there while the shape becomes unrecognisable. Measured drift over 100 time units — ten to sixteen pendulum periods, depending on the setting — is 18 parts per million at the gentlest end and never worse than ± 300 ppm anywhere on the slider — that is integrator and polygon-resolution error, not physics: the exact statement (1.3.46) has no error at all. **What no static figure shows.** That the distortion is unbounded and the measure is exactly fixed, simultaneously, in the same object.

△ WHY THIS ISN'T OBVIOUS

Liouville's theorem seems to forbid the second law of thermodynamics. Entropy is supposed to increase; entropy is a measure of the phase-space volume compatible with what you know; but the volume is exactly constant. Something has to give.

Look again at the figure. After a few periods the blob is a filament winding through a large region. Its *area* is what it was. But ask a different question: how much of the plane is within ε of some point of the blob, for a small but nonzero ε ? That number grows enormously, because a long thin filament has a huge neighbourhood. Any description of the system with finite resolution — any thermodynamic description, since you cannot record 10^{23} coordinates — sees the coarse-grained blob, and the coarse-grained volume is *not* conserved. It increases, monotonically in practice, and its logarithm is the entropy.

So the arrow of time is not in the dynamics. The dynamics is exactly reversible and exactly volume-preserving, forwards and backwards, and the figure runs equally happily either way. The arrow is in the **coarse-graining**: in the fact that stirring makes a distribution *look* uniform at any finite resolution, and that we are obliged to describe systems at finite resolution. Gibbs's own image was a drop of ink stirred into water — the ink occupies the same volume it always did, but you will never unstir it, because unstirring requires knowing where every filament went.

Two consequences worth having. First, "entropy increases" is a statement about our description, not a term in any equation of motion; there is no $+\sigma$ anywhere in (1.3.24). Second, ▀ quoted: Poincaré's recurrence theorem says that a Hamiltonian system in a bounded phase-space region must return arbitrarily close to its initial state, and the proof is Liouville — if it never returned, the images of a small blob would be disjoint forever and their total volume would exceed the finite volume available. The recurrence times are absurd (for a mole of gas, exponentially large in 10^{23}), which is why this does not overturn thermodynamics, but it is not a loophole, it is a theorem, and it exists because of the three lines above.

6 · Poisson brackets

We now extract from Hamilton's equations the structure that will survive quantisation. Everything in this section is elementary; its importance is entirely in what it becomes.

6.1 · The equation of motion for any observable whatsoever

An **observable** is any function $f(q, p, t)$ on phase space: the energy, a component of angular momentum, the distance between two particles, the electric dipole moment, anything you could in principle measure. Ask how it changes along a trajectory. By the multivariable chain rule (Chapter 0.6),

$$\frac{df}{dt} = \sum_i \left(\frac{\partial f}{\partial q^i} \dot{q}^i + \frac{\partial f}{\partial p_i} \dot{p}_i \right) + \frac{\partial f}{\partial t}. \quad (1.3.47)$$

Substitute Hamilton's equations (1.3.24) for $\dot{q}$ and $\dot{p}$:

$$\frac{df}{dt} = \sum_i \left(\frac{\partial f}{\partial q^i} \frac{\partial H}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial H}{\partial q^i} \right) + \frac{\partial f}{\partial t}. \quad (1.3.48)$$

The bracketed combination has appeared without being asked for, so give it a name. For any two functions f, g on phase space define the **Poisson bracket**

$$\{f, g\} \equiv \sum_i \left(\frac{\partial f}{\partial q^i} \frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial g}{\partial q^i} \right) = (\nabla f)^\top \Omega (\nabla g), \quad (1.3.49)$$

the second form using the symplectic matrix of (1.3.31) — check the block structure and you will find it reproduces the sum exactly. Then (1.3.48) reads

$$\frac{df}{dt} = \{f, H\} + \frac{\partial f}{\partial t}. \quad (1.3.50)$$

This is not a formula about a particular system. It is **the equation of motion for every observable of every Hamiltonian system**, and it contains Hamilton's equations as the special cases $f = q^i$ and $f = p_i$ (verify: $\{q^i, H\} = \partial H / \partial p_i$ and $\{p_i, H\} = -\partial H / \partial q^i$ ✓). It also contains, immediately, the master conservation law: if f has no explicit time dependence, then

$$f \text{ is conserved} \iff \{f, H\} = 0. \quad (1.3.51)$$

Conservation has become an algebraic condition. You no longer solve the equations of motion to find out whether something is conserved; you compute one bracket. Taking $f = H$ gives $\{H, H\} = 0$ trivially by antisymmetry, recovering (1.3.25).

6.2 · The fundamental brackets

Compute the brackets of the coordinates with each other. For $\{q^i, p_j\}$, note that $\partial q^i / \partial q^k = \delta_k^i$, $\partial q^i / \partial p_k = 0$, $\partial p_j / \partial p_k = \delta_{jk}$, $\partial p_j / \partial q^k = 0$, so only the first term of (1.3.49) survives:

$$\{q^i, p_j\} = \sum_k \delta_k^i \delta_{jk} = \delta_j^i, \quad \{q^i, q^j\} = 0, \quad \{p_i, p_j\} = 0. \quad (1.3.52)$$

The last two vanish because in each case one of the two factors in every term is a derivative of a q with respect to a p , or vice versa. These three lines are the **canonical commutation relations** of classical mechanics. Every coordinate

commutes with every other coordinate; every momentum with every other momentum; and a coordinate fails to commute with its own conjugate momentum, by exactly 1.

6.3 · The algebra

The bracket has four properties, and they are the properties, not incidental features.

Bilinearity. $\{af + bg, h\} = a\{f, h\} + b\{g, h\}$ for constants a, b , and likewise in the second slot. Immediate from (1.3.49), since differentiation is linear.

Antisymmetry. $\{f, g\} = -\{g, f\}$, and hence $\{f, f\} = 0$. Immediate from $\Omega^T = -\Omega$: swapping f and g in $(\nabla f)^T \Omega \nabla g$ transposes a number, which does nothing, and transposing the middle factor flips the sign.

Leibniz. $\{f, gh\} = \{f, g\}h + g\{f, h\}$. Immediate from the product rule, since every term of (1.3.49) differentiates g exactly once. This says $\{f, \cdot\}$ acts as a *derivation* — it behaves like a directional derivative, which §7 will explain by exhibiting the direction.

The Jacobi identity.

$$\{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\} = 0. \quad (1.3.53)$$

This one is not obvious and is proved in the grind box. It is the compatibility condition that makes everything hang together: bilinear, antisymmetric, and Jacobi is the definition of a **Lie algebra**, the structure Chapter 6.1 builds a whole part of the book on. One concrete consequence, available now: if f and g are both conserved, so is $\{f, g\}$. Proof: put $h = H$ in (1.3.53), use $\{f, H\} = \{g, H\} = 0$, and the first two terms die, leaving $\{H, \{f, g\}\} = 0$. This is **Poisson's theorem**, and it is a machine for manufacturing conservation laws from ones you already have — occasionally producing something new, usually producing something you knew, and either way not costing you an integration.

▼ Grind box — proving the Jacobi identity, and why no second derivatives can survive

Write everything in the compact notation of (1.3.49). Let $z^a, a = 1, \dots, 2n$, be the phase-space coordinates, $\partial_a = \partial/\partial z^a$, and Ω^{ab} the constant antisymmetric matrix of (1.3.31), so that

$$\{f, g\} = \Omega^{ab} \partial_a f \partial_b g$$

with repeated indices summed. The two facts we will use are that Ω^{ab} is *constant* (so it passes through derivatives) and *antisymmetric*.

The counting argument first, because it tells you why the identity is true. Each term of (1.3.53) is a bracket of a bracket, so it contains exactly two derivatives distributed over three functions; every term therefore contains a *second* derivative of exactly one of f, g, h and first derivatives of the other two. The claim is that the second-derivative terms cancel among themselves — separately for f , for g and for h — and that once they are gone nothing is left, since every term has a second derivative somewhere. So it suffices to check one of the three cancellations; the other two follow by relabelling.

The cancellation. Expand the first and third terms by the product rule:

$$\begin{aligned}\{f, \{g, h\}\} &= \Omega^{ab} \partial_a f \partial_b (\Omega^{cd} \partial_c g \partial_d h) \\ &= \Omega^{ab} \Omega^{cd} [\partial_a f \partial_b \partial_c g \partial_d h + \partial_a f \partial_c g \partial_b \partial_d h], \\ \{h, \{f, g\}\} &= \Omega^{ab} \Omega^{cd} [\partial_a h \partial_b \partial_c f \partial_d g + \partial_a h \partial_c f \partial_b \partial_d g].\end{aligned}$$

Collect the two terms carrying second derivatives of g : they are

$$T_1 = \Omega^{ab} \Omega^{cd} \partial_a f \partial_d h \partial_b \partial_c g, \quad T_2 = \Omega^{ab} \Omega^{cd} \partial_a h \partial_c f \partial_b \partial_d g.$$

(The middle term of (1.3.53), $\{g, \{h, f\}\}$, contains second derivatives of h and of f only, so it contributes nothing here.) Now relabel the dummy indices in T_2 by the substitution $a \rightarrow d, b \rightarrow c, c \rightarrow a, d \rightarrow b$:

$$T_2 = \Omega^{dc} \Omega^{ab} \partial_d h \partial_a f \partial_c \partial_b g = -\Omega^{cd} \Omega^{ab} \partial_a f \partial_d h \partial_b \partial_c g = -T_1,$$

using $\Omega^{dc} = -\Omega^{cd}$ once and the symmetry $\partial_c \partial_b g = \partial_b \partial_c g$ (Clairaut) once. So $T_1 + T_2 = 0$. By the cyclic symmetry of (1.3.53), the second-derivative-of- f terms and the second-derivative-of- h terms cancel by the identical relabelling. Nothing remains. ■

Note what the proof used: antisymmetry of Ω , constancy of Ω , and equality of mixed partials. Nothing else, and in particular nothing about mechanics. The identity is a fact about any constant antisymmetric bilinear pairing of gradients.

Verification, since a proof by index gymnastics deserves a check. Taking the deliberately unstructured functions $f = x^2 p_y + z p_x p_y + p_z \sin y, g = y p_x^2 + x z p_z + p_y \cos z, h = z^3 p_x + y p_y p_z + x y z$ on a three-degree-of-freedom phase space, symbolic differentiation gives a Jacobi residual of exactly 0, and the Leibniz residual $\{f, gh\} - \{f, g\}h - g\{f, h\}$ is exactly 0 as well.

6.4 · The payoff, and the reason Part I exists

Here is what the last three pages were for.

CANONICAL QUANTIZATION, IN ONE LINE

Chapter 4.6 will take the classical structure you now own, replace observables $f(q, p)$ by self-adjoint operators $\hat{f}$ on a Hilbert space, and make the single substitution

$$\{f, g\} \longmapsto \frac{1}{i\hbar} [\hat{f}, \hat{g}], \quad [\hat{f}, \hat{g}] \equiv \hat{f}\hat{g} - \hat{g}\hat{f}.$$

Nothing else changes. Every equation of this section survives verbatim.

Watch what that does. The fundamental brackets (1.3.52) become

$$[\hat{q}^i, \hat{p}_j] = i\hbar \delta_j^i, \quad [\hat{q}^i, \hat{q}^j] = 0, \quad [\hat{p}_i, \hat{p}_j] = 0, \quad (1.3.54)$$

which is the foundational relation of quantum mechanics, from which the uncertainty principle follows in three lines (Chapter 4.6) and the operator $\hat{p} = -i\hbar \partial / \partial q$ is the standard realisation. The equation of motion (1.3.50) becomes

$$\frac{d\hat{f}}{dt} = \frac{1}{i\hbar} [\hat{f}, \hat{H}] + \frac{\partial \hat{f}}{\partial t}, \quad (1.3.55)$$

the **Heisenberg equation of motion**, which is the whole of quantum dynamics in the Heisenberg picture. And the conservation criterion (1.3.51) becomes " $\hat{f}$ is conserved if and only if it commutes with the Hamiltonian" — the statement that gets used, without comment, in every quantum mechanics course from the second week onward.

The bracket properties go across too, and they are exactly the properties a commutator has. Bilinearity: obvious. Antisymmetry: $[\hat{f}, \hat{g}] = -[\hat{g}, \hat{f}]$ by inspection. Leibniz: $[\hat{f}, \hat{g}\hat{h}] = [\hat{f}, \hat{g}]\hat{h} + \hat{g}[\hat{f}, \hat{h}]$, which you can verify in one line by inserting and cancelling $\hat{g}\hat{f}\hat{h}$. Jacobi: holds for commutators identically, by expanding all twelve products and watching them cancel in pairs. **The classical Poisson algebra and the quantum commutator algebra are the same Lie algebra.** That is the sense in which the substitution "changes nothing else".

So the answer to "why did we spend a chapter on a change of variables" is: because the change of variables is the thing that quantises. You cannot quantise the Lagrangian formulation directly — there is no bracket there, only a variational principle — which is why Feynman had to invent an entirely different route (the path integral, Chapter 5.8) to quantise from an action. Canonical quantisation is a one-line substitution into a structure you now have.

△ WHY THIS ISN'T OBVIOUS

"Replace brackets by commutators" is a *correspondence*, not a theorem, and it is not quite consistent.

■ Quoted: the Groenewold–van Hove theorem proves that there is **no** map from classical observables to operators that sends $\{f, g\}$ to $[\hat{f}, \hat{g}]/i\hbar$ for *all* f, g while also sending $q \mapsto \hat{q}$ and $p \mapsto \hat{p}$. The obstruction shows up at cubic order: $\{q^2, p^2\}$ and the corresponding commutator disagree once you decide how to order $\hat{q}$'s and $\hat{p}$'s. So the substitution is exact for observables at most quadratic in q and p — which covers the harmonic oscillator, free particles, angular momentum, and therefore most of Chapter 4 — and needs a convention (an ordering prescription) beyond that.

This is not a scandal; it is a statement that classical mechanics does not uniquely determine its quantum parent, which we should have expected, since the arrow of explanation runs the other way. Quantum mechanics is the theory; classical mechanics is its $\hbar \rightarrow 0$ shadow, and shadows do not determine the objects that cast them. Keep the correspondence as what it is: an extremely good guess, exact where it matters most, and the reason the structure of this chapter is worth owning.

7 · Symmetries generate motion

This is the deepest section in the chapter, and it is short. It answers a question you may not have thought to ask: *why* should conserved quantities and symmetries have anything to do with one another?

Chapter 1.4 will prove Noether's theorem, which says: every continuous symmetry of the action implies a conserved quantity. That is a one-way street, and it leaves the conserved quantity looking like a by-product — something the symmetry *emits*. The Hamiltonian picture reveals that the relationship is far tighter than that.

7.1 · Every observable generates a flow

Take any function $G(q, p)$ on phase space and use it in place of H in Hamilton's equations. That is, define an infinitesimal change of every observable f by

$$\delta f = \epsilon \{f, G\}, \quad \epsilon \text{ infinitesimal.} \quad (1.3.56)$$

Applied to the coordinates themselves this reads $\delta q^i = \epsilon \partial G / \partial p_i$ and $\delta p_i = -\epsilon \partial G / \partial q^i$ — the same structure as (1.3.24) with ϵ playing the role of a small time and G the role of the Hamiltonian. So G defines a flow on phase space, and (1.3.56) is how any observable changes along it. We say G **generates** the transformation.

Before doing examples, note that this transformation is automatically canonical — it preserves the fundamental brackets. To first order in ϵ , using bilinearity and then Clairaut,

$$\begin{aligned} \{q + \delta q, p + \delta p\} &= \{q, p\} + \epsilon \left\{ \frac{\partial G}{\partial p}, p \right\} - \epsilon \left\{ q, \frac{\partial G}{\partial q} \right\} + O(\epsilon^2) \\ &= 1 + \epsilon \frac{\partial^2 G}{\partial q \partial p} - \epsilon \frac{\partial^2 G}{\partial p \partial q} + O(\epsilon^2) = 1 + O(\epsilon^2). \end{aligned} \quad (1.3.57)$$

Mixed partials again — the third time in this chapter, after (1.3.45) and the Jacobi identity. Whatever G you pick, the flow it generates is a symmetry of the symplectic structure. Now the three cases that matter.

7.2 · The three canonical examples

The Hamiltonian generates time translation. Take $G = H$ and $\epsilon = dt$. Then (1.3.56) is $\delta f = dt \{f, H\} = dt \frac{df}{dt}$ for any f without explicit time dependence, by (1.3.50). So the flow generated by H is *the dynamics itself*: evolving the system forward by dt and applying the transformation generated by H are the same operation. The Hamiltonian is not merely the energy; it is the generator of translations in time.

Momentum generates spatial translation. Take $G = p_x$, the momentum conjugate to x . Then for any f ,

$$\{f, p_x\} = \frac{\partial f}{\partial x} \frac{\partial p_x}{\partial p_x} = \frac{\partial f}{\partial x} \implies \delta f = \epsilon \frac{\partial f}{\partial x}, \quad (1.3.58)$$

every other term of (1.3.49) vanishing because p_x depends on nothing else. And $\epsilon \partial f / \partial x$ is precisely the first-order change in f when you move the system a distance ϵ in the x -direction — Chapter 0.1's linear approximation. So p_x

generates translations along x . In particular $\delta x = \epsilon$ and $\delta p = 0$: the momentum shifts nothing and the position shifts by ϵ , which is exactly what a translation does.

Angular momentum generates rotation. Take $G = L_z = xp_y - yp_x$. Compute the brackets with the four relevant coordinates directly from (1.3.49):

$$\begin{aligned} \{x, L_z\} &= \frac{\partial L_z}{\partial p_x} = -y, & \{y, L_z\} &= \frac{\partial L_z}{\partial p_y} = +x, \\ \{p_x, L_z\} &= -\frac{\partial L_z}{\partial x} = -p_y, & \{p_y, L_z\} &= -\frac{\partial L_z}{\partial y} = +p_x. \end{aligned} \quad (1.3.59)$$

So (1.3.56) gives $\delta x = -\epsilon y$, $\delta y = +\epsilon x$, and identically $\delta p_x = -\epsilon p_y$, $\delta p_y = +\epsilon p_x$. Written out,

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} 1 & -\epsilon \\ \epsilon & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}, \quad (1.3.60)$$

which is a rotation by the angle ϵ about the z -axis, to first order — and the momenta rotate the same way, as vectors must. L_z generates rotations about z . Nothing was assumed about the system; this is a statement about the function $xp_y - yp_x$ and the bracket, valid for any Hamiltonian whatsoever.

7.3 · The two-way street

Now put the pieces together, and note that everything needed is antisymmetry.

SYMMETRY AND CONSERVATION ARE THE SAME STATEMENT, READ IN TWO DIRECTIONS

Let G have no explicit time dependence. Then

$$\underbrace{\frac{dG}{dt} = \{G, H\}}_{G \text{ is conserved iff this vanishes}} \quad \text{and} \quad \underbrace{\delta H = \epsilon \{H, G\} = -\epsilon \{G, H\}}_{H \text{ is invariant under } G\text{'s flow iff this vanishes}}$$

are the same quantity up to sign. Therefore

$$G \text{ is conserved} \iff \{G, H\} = 0 \iff H \text{ is invariant under the flow generated by } G.$$

Read it left to right and it is the converse of Noether's theorem. Read it right to left and it is Noether's theorem. And read as a single line it says something stronger than either: the conserved quantity is not merely *associated* with the symmetry, and is not a by-product of it. **The conserved quantity is the generator of the symmetry.** They are one object. Momentum is spatial translation; angular momentum is rotation; energy is time translation. The bracket $\{G, H\}$ is simultaneously "the rate of change of G " and "the change in H under G ", and there is nothing else to say.

This is the single most portable idea in the book, and here is where it goes.

- **Chapter 1.4** proves Noether's theorem for a general Lagrangian, including the field-theory version where the conserved quantity becomes a conserved *current*. This section is the Hamiltonian shadow of it, and it is the direction Noether does not give you.
- **Chapter 4.2** repeats every word with operators. There, an observable $\hat{G}$ generates the one-parameter family of unitary transformations $e^{-i\epsilon\hat{G}/\hbar}$, and differentiating that at $\epsilon = 0$ gives $\delta\hat{f} = \frac{i\epsilon}{\hbar}[\hat{G}, \hat{f}]$, which is (1.3.56) under §6.4's substitution. "Observables generate unitary transformations" is this section, quantised. It is why $\hat{p} = -i\hbar\partial_x$ — the generator of translations — and why $\hat{H}$ appears in the exponent of the time-evolution operator.
- **Chapters 6.1 and 6.2** take the set of generators seriously as an object in its own right. The brackets of generators with each other close into an algebra — Problem 2 computes $\{L_i, L_j\} = \epsilon_{ijk}L_k$ and finds $\mathfrak{su}(2)$ arriving classically — and a Lie algebra with its representations is what a "symmetry group" concretely is. Chapter 6.3 then makes the parameter ϵ depend on position, discovers that this forces a new field into existence, and that field is the photon. The entire gauge principle is this section with $\epsilon \rightarrow \epsilon(x)$.

8 · Canonical transformations and Hamilton–Jacobi

This section is a signpost, not a development. Two ideas, stated precisely enough to be usable, with the machinery left for the chapters that need it.

8.1 · Which changes of variable are allowed

Chapter 1.2 §7 proved that the Euler–Lagrange equations keep their form under *any* change of coordinates. Hamilton's equations are more demanding, because they involve the momenta, and the momenta are not free to be relabelled independently: they were defined by (1.3.11).

Call a transformation $(q, p) \rightarrow (Q, P)$ **canonical** if it preserves the fundamental brackets,

$$\{Q^i, P_j\} = \delta_j^i, \quad \{Q^i, Q^j\} = 0, \quad \{P_i, P_j\} = 0, \quad (1.3.61)$$

with the brackets still computed with respect to the *old* variables. Equivalently, in the notation of §3.3: if $M^a_b = \partial z'^a / \partial z^b$ is the Jacobian matrix of the transformation, (1.3.61) is precisely the statement

$$M \Omega M^T = \Omega, \quad (1.3.62)$$

which is the definition of a **symplectic matrix**. Such transformations preserve Hamilton's equations (the new equations are $\dot{Q} = \partial K / \partial P$, $\dot{P} = -\partial K / \partial Q$ for a suitable K), preserve every Poisson bracket whatsoever — since (1.3.49) is built from Ω — and, by taking determinants of (1.3.62), satisfy $(\det M)^2 = 1$. ▀ The sharper statement that $\det M = +1$ always, never -1 , is quoted; it follows from a Pfaffian argument. So canonical transformations preserve phase-space volume, and Liouville's theorem is the special case in which the transformation is "evolve for a time t " — the flow map is canonical, which Problem 3 asks you to check.

Generating functions, in one paragraph. The practical way to construct canonical transformations is to notice that the phase-space action (1.3.26) may differ between old and new variables by a total time derivative without changing the equations of motion (Chapter 1.2, Problem 4). Demand

$$\sum_i p_i \dot{q}^i - H = \sum_i P_i \dot{Q}^i - K + \frac{dF}{dt}, \quad (1.3.63)$$

and let $F = F_1(q, Q, t)$. Expanding dF_1/dt by the chain rule and matching the coefficients of the independent quantities $\dot{q}^i$, $\dot{Q}^i$ and 1 gives

$$p_i = \frac{\partial F_1}{\partial q^i}, \quad P_i = -\frac{\partial F_1}{\partial Q^i}, \quad K = H + \frac{\partial F_1}{\partial t}. \quad (1.3.64)$$

Any F_1 you write down generates a canonical transformation. Three cousins $F_2(q, P, t)$, $F_3(p, Q, t)$, $F_4(p, P, t)$ follow by Legendre-transforming F_1 in one or both arguments — §1 again, in yet another costume. The grind box works the most useful one.

8.2 · Hamilton–Jacobi

Push the idea to its limit. Suppose you could find a canonical transformation making the new Hamiltonian K *identically zero*. Then $\dot{Q} = \partial K / \partial P = 0$ and $\dot{P} = -\partial K / \partial Q = 0$: all $2n$ new variables are constants, and the system is completely solved. From (1.3.64) with the generating function called S instead of F_1 , $K = 0$ means $H + \partial S / \partial t = 0$ with $p_i = \partial S / \partial q^i$ substituted into H :

$$\boxed{\frac{\partial S}{\partial t} + H\left(q^1, \dots, q^n, \frac{\partial S}{\partial q^1}, \dots, \frac{\partial S}{\partial q^n}, t\right) = 0.} \quad (1.3.65)$$

This is the **Hamilton–Jacobi equation**: one first-order partial differential equation for one function $S(q, t)$, entirely equivalent to the $2n$ ordinary differential equations we started with. It is rarely the easiest way to solve a mechanics problem. It is on this page for two reasons.

First, S is the action. Not a function that resembles the action — the action. Compute its total time derivative along a trajectory, using $\partial S / \partial q^i = p_i$ and (1.3.65):

$$\frac{dS}{dt} = \frac{\partial S}{\partial t} + \sum_i \frac{\partial S}{\partial q^i} \dot{q}^i = -H + \sum_i p_i \dot{q}^i = L, \quad (1.3.66)$$

so $S = \int L dt$ along the path. The generating function that trivialises the dynamics is Chapter 1.2's action regarded as a function of the endpoint. That is why the letter is the same, and it closes a loop opened two chapters ago.

Second, it is the classical limit of the Schrödinger equation. ■ Quoted, with the derivation deferred to Chapter 4.6. Substitute the ansatz

$$\psi(\mathbf{x}, t) = A(\mathbf{x}, t) e^{iS(\mathbf{x}, t)/\hbar} \quad (1.3.67)$$

into $i\hbar \partial_t \psi = -\frac{\hbar^2}{2m} \nabla^2 \psi + V \psi$ and separate the real and imaginary parts. The real part, at leading order in $\hbar$, is

$$\frac{\partial S}{\partial t} + \frac{|\nabla S|^2}{2m} + V = 0, \quad (1.3.68)$$

which is exactly (1.3.65) for $H = \mathbf{p}^2/2m + V$ with $\mathbf{p} = \nabla S$; the terms that were dropped are down by one power of $\hbar$. (The imaginary part gives $\partial_t |A|^2 + \nabla \cdot (|A|^2 \nabla S/m) = 0$, which is Chapter 0.7's continuity equation for probability, with velocity $\nabla S/m$.) So the phase of a quantum wavefunction is the classical action divided by $\hbar$ — which is also what Chapter 1.2 §5 quoted about the path integral weight $e^{iS/\hbar}$, and what Chapter 5.8 will prove. Classical mechanics is the geometry of surfaces of constant phase, in exactly the way that ray optics is the geometry of surfaces of constant phase of a light wave. Hamilton knew this in 1834, which is ninety-two years before anyone wrote down a wave equation for matter.

▼ Grind box — the F_2 generating function, and the oscillator solved by Hamilton–Jacobi

The transform between generating functions. $F_1(q, Q)$ is awkward because the identity transformation is not expressible in it. Legendre-transform the Q argument: set $F_2(q, P, t) = F_1(q, Q, t) + \sum_i Q^i P_i$, treating $P_i = -\partial F_1/\partial Q^i$ as the conjugate slope. Then $dF_2 = \sum_i p_i dq^i + \sum_i Q^i dP_i + (\partial F_1/\partial t)dt$, giving

$$p_i = \frac{\partial F_2}{\partial q^i}, \quad Q^i = \frac{\partial F_2}{\partial P_i}, \quad K = H + \frac{\partial F_2}{\partial t}.$$

Now $F_2 = \sum_i q^i P_i$ gives $p_i = P_i$ and $Q^i = q^i$ — the identity, as required. This is the version used in practice, and the one Hamilton–Jacobi uses.

The oscillator, solved from scratch. Take $H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2$. Since H has no explicit t , look for a separated solution $S(q, t) = W(q) - Et$, so that $\partial S/\partial t = -E$ and (1.3.65) becomes

$$\frac{1}{2m} \left(\frac{dW}{dq} \right)^2 + \frac{1}{2}m\omega^2 q^2 = E \implies \frac{dW}{dq} = \sqrt{2mE - m^2\omega^2 q^2}.$$

So $W(q) = \int \sqrt{2mE - m^2\omega^2 q^2} dq$, and $S = W(q) - Et$. Take the constant E to be the new momentum P . Then the new coordinate is

$$Q = \frac{\partial S}{\partial E} = \int \frac{m dq}{\sqrt{2mE - m^2\omega^2 q^2}} - t = \frac{1}{\omega} \arcsin \left(q \omega \sqrt{\frac{m}{2E}} \right) - t,$$

and Q is constant because $K = H + \partial S/\partial t = E - E = 0$. Setting $Q = -t_0$ and solving for q :

$$q(t) = \sqrt{\frac{2E}{m\omega^2}} \sin(\omega(t - t_0)).$$

The oscillator solution, with the amplitude expressed through the energy exactly as Chapter 0.8 had it. Note what the method did: it turned solving a differential equation into evaluating an integral, and the integral was the one whose derivative is the period — (1.3.41) again, since $\oint dW = \oint p dq = \mathcal{A}$.

And the action variable. Instead of E , take the new momentum to be $J = \mathcal{A}/2\pi = E/\omega$. Then $K = \omega J$ and Hamilton's equations give $\dot{\vartheta} = \partial K / \partial J = \omega$ and $\dot{J} = 0$: an angle advancing uniformly and a constant. Every one-dimensional bound system can be brought to this form, which is the content of "action-angle variables" and the starting point for perturbation theory in celestial mechanics — and, ► quoted forward, the starting point for the KAM theorem, which says what survives when you switch a small perturbation on and is the reason the solar system's stability is a hard question rather than an obvious one.

9 • Worked examples

WORKED EXAMPLE 1 — THE CHARGED PARTICLE, AND MINIMAL COUPLING THREE PARTS EARLY

Take the Lagrangian (1.3.12) for a particle of mass m and charge e in given potentials $\phi(\mathbf{x}, t)$ and $\mathbf{A}(\mathbf{x}, t)$:

$$L = \frac{1}{2}m \mathbf{v} \cdot \mathbf{v} - e\phi + e \mathbf{v} \cdot \mathbf{A}.$$

(a) It is the right Lagrangian. Before transforming it, check it. The canonical momentum is (1.3.13), $p_i = mv_i + eA_i$, so the Euler-Lagrange equation $\dot{p}_i = \partial L / \partial x^i$ reads

$$\frac{d}{dt}(mv_i + eA_i) = -e \frac{\partial \phi}{\partial x^i} + e v^j \frac{\partial A_j}{\partial x^i}.$$

The total time derivative of A_i along the trajectory is $dA_i/dt = \partial_t A_i + v^j \partial_j A_i$ (chain rule, Chapter 0.6). Move it to the right:

$$m\dot{v}_i = e \underbrace{\left(-\frac{\partial \phi}{\partial x^i} - \frac{\partial A_i}{\partial t} \right)}_{= E_i} + e v^j \left(\frac{\partial A_j}{\partial x^i} - \frac{\partial A_i}{\partial x^j} \right).$$

The first bracket is the electric field. The second is the antisymmetric part of the Jacobian of $\mathbf{A}$, which Chapter 0.7 §4.3 identified with the curl: writing $\mathbf{B} = \nabla \times \mathbf{A}$, the identity $v^j (\partial_i A_j - \partial_j A_i) = (\mathbf{v} \times \mathbf{B})_i$ holds component by component. So

$$m\dot{\mathbf{v}} = e(\mathbf{E} + \mathbf{v} \times \mathbf{B}),$$

the Lorentz force. ✓ (Verified symbolically for all three components.) The Lagrangian is correct, and note the structure: the $\mathbf{v} \cdot \mathbf{A}$ term is linear in velocity, so it produces a force linear in velocity, which is what a magnetic force is.

(b) The Hamiltonian. By §2.3's grind box, $H = L_2 - L_0 = \frac{1}{2}mv^2 + e\phi$. But H must be expressed in the momenta, and from (1.3.13), $m\mathbf{v} = \mathbf{p} - e\mathbf{A}$. Substituting,

$$H = \frac{(\mathbf{p} - e\mathbf{A})^2}{2m} + e\phi.$$

Look at what happened. The vector potential vanished from H when written in velocities and reappeared inside a square when written in momenta. The magnetic field does no work — that is the first statement — and yet it is not absent from the dynamics — that is the second. Both are true, and the Hamiltonian formalism is what makes them compatible.

(c) It reproduces the same physics. Hamilton's equations give $\dot{x}^i = \partial H / \partial p_i = (p^i - eA^i)/m$, which is $m\mathbf{v} = \mathbf{p} - e\mathbf{A}$ again, and $\dot{p}_i = -\partial H / \partial x^i$, which after substituting the first equation reduces to the Lorentz force. (Verified symbolically; the residual is exactly zero once $\dot{\mathbf{x}} = (\mathbf{p} - e\mathbf{A})/m$ is used.)

(d) Why this box exists. The boxed Hamiltonian is the object Chapter 6.3 quantises. The rule there — **minimal coupling** — will be stated as: to couple a charged particle to electromagnetism, replace

$$\mathbf{p} \longrightarrow \mathbf{p} - e\mathbf{A}, \quad \text{equivalently} \quad -i\hbar\nabla \longrightarrow -i\hbar\nabla - e\mathbf{A},$$

everywhere in the free Hamiltonian, and nothing else. Chapter 6.3 derives that rule from the requirement that the theory be invariant under a phase rotation $\psi \rightarrow e^{ie\chi/\hbar}\psi$ whose parameter varies from point to point, and then repeats the derivation with the phase replaced by an $SU(3)$ matrix to obtain the strong force. All of it is the substitution above. You have now derived that substitution, from a Legendre transform, in classical mechanics, with no quantum mechanics and no gauge theory in sight. When Chapter 6.3 tells you that the covariant derivative $D_\mu = \partial_\mu - ieA_\mu$ is forced, you will recognise it as this worked example wearing indices.

WORKED EXAMPLE 2 — THE PENDULUM PHASE PORTRAIT, WITH EVERY NUMBER COMPUTED

Take (1.3.36) with $m = \ell = 1$ and $g = \omega_0^2$, so $H = \frac{1}{2}p^2 + \omega_0^2(1 - \cos q)$. We work out the complete structure of the phase portrait — the thing the interactive draws underneath the blob.

(a) Fixed points. $\dot{q} = p = 0$ and $\dot{p} = -\omega_0^2 \sin q = 0$ require $p = 0$ and $q = 0$ or $q = \pi$ (modulo 2π). Two fixed points per period, and no others.

(b) Their character. Linearise the flow at each. Writing the state as $z = (q, p)$ and $\dot{z} = \mathbf{v}(z)$, the Jacobian of $\mathbf{v}$ is

$$D\mathbf{v} = \begin{pmatrix} \partial_q \dot{q} & \partial_p \dot{q} \\ \partial_q \dot{p} & \partial_p \dot{p} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\omega_0^2 \cos q & 0 \end{pmatrix}.$$

Its trace is zero everywhere — which is Liouville's theorem (1.3.45) seen locally, since the trace of the Jacobian is the divergence. Its determinant is $\omega_0^2 \cos q$, so the eigenvalues satisfy $\lambda^2 = -\omega_0^2 \cos q$:

FIXED POINT	λ^2	EIGENVALUES	TYPE
$q = 0$ (hanging)	$-\omega_0^2$	$\pm i\omega_0$	centre — closed orbits
$q = \pi$ (inverted)	$+\omega_0^2$	$\pm\omega_0$	saddle — hyperbolic

Because the trace vanishes, the eigenvalues always come in $\pm$ pairs: a Hamiltonian system in one degree of freedom can only have centres and saddles, never spirals or nodes. Spiralling in would contract volume, and Liouville forbids it. Chapter 0.8's classification of damped oscillators had four cases; here two of them are simply illegal.

(c) The separatrix energy, exactly. The saddle sits at $(q, p) = (\pi, 0)$, so its energy is $H(\pi, 0) = 0 + \omega_0^2(1 - \cos \pi) = 2\omega_0^2$ — restoring units, $H_{\text{sep}} = 2mgl$, which is (1.3.37) and which is exactly the potential energy of a bob lifted from the bottom to the top: $mg \cdot 2\ell$. The separatrix is the level set through the saddle, given by (1.3.38), $p = \pm 2\omega_0 \cos(q/2)$, which is the orange curve in the figure. Its maximum height is $p = 2\omega_0$ at $q = 0$: the pendulum needs exactly the angular velocity $2\omega_0$ at the bottom to *just* reach the top.

(d) Time on the separatrix. On it, $\dot{q} = p = 2\omega_0 \cos(q/2)$, which separates:

$$\omega_0 t = \int \frac{dq}{2 \cos(q/2)} = \ln \left[\tan\left(\frac{q}{4} + \frac{\pi}{4}\right) \right] + \text{const},$$

which diverges logarithmically as $q \rightarrow \pi$. Inverting, $q(t) = 4 \arctan(e^{\omega_0 t}) - \pi$, the famous *homoclinic orbit*: it leaves the inverted position at $t = -\infty$ and returns to it at $t = +\infty$, taking infinite time at both ends and a finite time in between. It is a single trajectory whose past and future limit points are the same point, which is why it can appear to "cross itself" at the saddle without violating §4.1.

(e) Why this makes the interactive violent. Just inside the separatrix, the period is $T(E) = 4K(E/2\omega_0^2)/\omega_0$ in the elliptic-integral notation of Chapter 0.8, and K diverges logarithmically as its argument approaches 1. At $\omega_0 = 1$: $T = 6.4498$ at $E = 0.2$, $T = 8.3015$ at $E = 1.4$, $T = 11.6333$ at $E = 1.9$, and $T \rightarrow \infty$ at $E = 2$. A blob of radius 0.14 centred at $p = 1.82$ spans momenta from 1.68 to

1.96, hence energies from 1.4112 to 1.9208, hence periods from 8.3348 to 12.0839 — a spread of 45%. After ten circuits the fast edge has lapped the slow edge several times, and the blob is a filament. Its area, by (1.3.46), has not moved.

10 · Your turn

PROBLEM 1 — LEGENDRE-TRANSFORM A LAGRANGIAN, AND CHECK THE INVOLUTION

A particle of mass m moves in a plane under a central potential $V(r)$. In polar coordinates, $L = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) - V(r)$ (Chapter 1.2 §7.1). (a) Find both canonical momenta and construct H . (b) Write down Hamilton's equations and identify the conserved quantities without solving anything. (c) Verify explicitly that Legendre-transforming H back with respect to the momenta returns L .

– Solution

(a) The momenta are

$$p_r = \frac{\partial L}{\partial \dot{r}} = m\dot{r}, \quad p_\theta = \frac{\partial L}{\partial \dot{\theta}} = mr^2\dot{\theta}.$$

Note p_θ is the angular momentum, and it is not $m \times$ (a velocity) — it carries an extra factor of r^2 , which is §2.1's warning made concrete. Inverting, $\dot{r} = p_r/m$ and $\dot{\theta} = p_\theta/mr^2$. Both hypotheses of §2.3 hold (T is a homogeneous quadratic, V is velocity-independent), so $H = T + V$; doing it by the definition anyway,

$$\begin{aligned} H &= p_r\dot{r} + p_\theta\dot{\theta} - L \\ &= \frac{p_r^2}{m} + \frac{p_\theta^2}{mr^2} - \left[\frac{p_r^2}{2m} + \frac{p_\theta^2}{2mr^2} - V(r) \right] = \frac{p_r^2}{2m} + \frac{p_\theta^2}{2mr^2} + V(r). \end{aligned}$$

(b) Hamilton's equations:

$$\dot{r} = \frac{p_r}{m}, \quad \dot{\theta} = \frac{p_\theta}{mr^2}, \quad \dot{p}_r = -\frac{\partial H}{\partial r} = \frac{p_\theta^2}{mr^3} - V'(r), \quad \dot{p}_\theta = -\frac{\partial H}{\partial \theta} = 0.$$

Two conservation laws, read straight off. θ does not appear in H , so p_θ is constant — angular momentum. And H has no explicit t , so H is constant — energy. That is two of the four constants of the motion, obtained by inspection rather than integration, which is the practical advantage of the Hamiltonian form. Substituting the constant $p_\theta = \ell$ into the r equations leaves a one-degree-of-freedom problem with effective potential $V_{\text{eff}}(r) = \ell^2/2mr^2 + V(r)$ — the centrifugal barrier, appearing here as an honest term in H rather than a fictitious force. With $V = -k/r$ this is the Kepler problem, and the phase portrait in (r, p_r) has exactly the structure of §4.3: a centre at the circular orbit, closed curves around it for bound elliptical orbits, and open curves for hyperbolic ones.

(c) **The involution.** Transform H back with respect to (p_r, p_θ) . The conjugate slopes are

$$\frac{\partial H}{\partial p_r} = \frac{p_r}{m} = \dot{r}, \quad \frac{\partial H}{\partial p_\theta} = \frac{p_\theta}{mr^2} = \dot{\theta},$$

reproducing (1.3.6): the transform of H has the velocities as its slopes, exactly mirroring $p = \partial L / \partial \dot{q}$. The transform is then

$$\dot{r} p_r + \dot{\theta} p_\theta - H = m\dot{r}^2 + mr^2\dot{\theta}^2 - \left[\frac{1}{2}m\dot{r}^2 + \frac{1}{2}mr^2\dot{\theta}^2 + V \right] = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) - V = L. \checkmark$$

Which is (1.3.8), verified on a real example. The formula $H = \sum p\dot{q} - L$ and the formula $L = \sum p\dot{q} - H$ are the same formula, and the apparent asymmetry between them is entirely in which variables you regard as independent.

With $\mathbf{L} = \mathbf{r} \times \mathbf{p}$, so that $L_x = yp_z - zp_y$ and cyclically, show that

$$\{L_i, L_j\} = \sum_k \epsilon_{ijk} L_k,$$

where ϵ_{ijk} is $+1$ for cyclic index orders, -1 for anticyclic, and 0 if any two indices agree. Then show $\{\mathbf{L}^2, L_z\} = 0$, and say what both results become in Chapter 4.7.

– Solution

The computation. Take $i = x, j = y$; the rest follow by cycling $x \rightarrow y \rightarrow z \rightarrow x$, which leaves the definitions invariant. The gradients of $L_x = yp_z - zp_y$ and $L_y = zp_x - xp_z$ are

$$\begin{aligned} \nabla_q L_x &= (0, p_z, -p_y), & \nabla_p L_x &= (0, -z, y), \\ \nabla_q L_y &= (-p_z, 0, p_x), & \nabla_p L_y &= (z, 0, -x). \end{aligned}$$

Now assemble (1.3.49) term by term over the three coordinates:

$$\begin{aligned} \{L_x, L_y\} &= \underbrace{[0 \cdot z - 0 \cdot (-p_z)]}_{x \text{ term}} + \underbrace{[p_z \cdot 0 - (-z) \cdot 0]}_{y \text{ term}} + \underbrace{[(-p_y)(-x) - y p_x]}_{z \text{ term}} \\ &= xp_y - yp_x = L_z. \end{aligned}$$

Only the z column survived, which is worth noticing: the bracket of two components of angular momentum "about" x and y is built entirely from the third direction. Cycling gives $\{L_y, L_z\} = L_x$ and $\{L_z, L_x\} = L_y$, and antisymmetry supplies the rest, so $\{L_i, L_j\} = \epsilon_{ijk} L_k$ in every case, including $\{L_i, L_i\} = 0$. (Verified symbolically.)

The Casimir. Use bilinearity and Leibniz on $\mathbf{L}^2 = L_x^2 + L_y^2 + L_z^2$:

$$\begin{aligned} \{\mathbf{L}^2, L_z\} &= 2L_x\{L_x, L_z\} + 2L_y\{L_y, L_z\} + 2L_z\underbrace{\{L_z, L_z\}}_{=0} \\ &= 2L_x(-L_y) + 2L_y(L_x) = 0. \checkmark \end{aligned}$$

The magnitude of the angular momentum commutes with every component, while no two components commute with each other.

What this is. The three functions L_x, L_y, L_z close under the bracket: the bracket of any two is a linear combination of the three. A set of generators with that property is a **Lie algebra**, and this particular one — three generators with structure constants ϵ_{ijk} — is $\mathfrak{su}(2)$, the same algebra as $\mathfrak{so}(3)$. You have just derived it inside classical mechanics, with no rotation matrices and no quantum mechanics.

By §6.4's substitution it becomes $[\hat{L}_i, \hat{L}_j] = i\hbar \epsilon_{ijk} \hat{L}_k$ and $[\hat{\mathbf{L}}^2, \hat{L}_z] = 0$ — the two relations from which Chapter 4.7 derives, with no further input, that angular momentum is quantised, that its magnitude

takes the values $\sqrt{j(j+1)}\hbar$, that j can be a half-integer, and hence that spin exists. Note what §7.2 already told you about these generators: they generate rotations. So "angular momentum is quantised" and "the rotation group has discrete representations" are the same fact, which is what Chapter 6.2 is about. Everything in that sentence was visible here, classically, as one bracket.

PROBLEM 3 — CANONICAL TRANSFORMATIONS HAVE UNIT JACOBIAN

(a) For one degree of freedom, show that a transformation $(q, p) \rightarrow (Q, P)$ satisfies $\{Q, P\} = 1$ if and only if its Jacobian determinant is 1, and explain how this makes Liouville's theorem a corollary. (b) Verify it on a concrete case: show that

$$q = \sqrt{\frac{2P}{m\omega}} \sin Q, \quad p = \sqrt{2Pm\omega} \cos Q$$

is canonical, and find what it does to the oscillator Hamiltonian.

– Solution

(a) With one degree of freedom the Jacobian matrix of the transformation is

$$M = \begin{pmatrix} \partial Q / \partial q & \partial Q / \partial p \\ \partial P / \partial q & \partial P / \partial p \end{pmatrix}, \quad \det M = \frac{\partial Q}{\partial q} \frac{\partial P}{\partial p} - \frac{\partial Q}{\partial p} \frac{\partial P}{\partial q}.$$

Compare with the definition (1.3.49):

$$\{Q, P\} = \frac{\partial Q}{\partial q} \frac{\partial P}{\partial p} - \frac{\partial Q}{\partial p} \frac{\partial P}{\partial q}.$$

They are literally the same expression. So $\{Q, P\} = 1 \iff \det M = 1$, with no computation at all. In n degrees of freedom the statement is (1.3.62), $M\Omega M^T = \Omega$; taking determinants and using $\det \Omega = 1 \neq 0$ gives $(\det M)^2 = 1$, and the sharper $\det M = +1$ is the fact quoted in §8.1.

Liouville as a corollary. Fix a time t and consider the map Φ_t that sends each initial condition (q_0, p_0) to its state at time t . This map is canonical: the brackets $\{q(t), p(t)\}$ computed with respect to (q_0, p_0) are preserved by the flow, because $\frac{d}{dt}\{q(t), p(t)\} = \{\{q, H\}, p\} + \{q, \{p, H\}\}$, which is zero by the Jacobi identity (1.3.53) applied to q, p and H together with $\{q, p\} = \text{constant}$. Hence $\det D\Phi_t = 1$, and by Chapter 0.6's change-of-variables formula the map preserves areas. That is Liouville's theorem again, derived algebraically rather than through the divergence — the two proofs are §5 and this, and it is worth having both, because §5's generalises to any divergence-free flow while this one explains why the phase-space *structure*, not merely the volume, is what is being conserved.

(b) Compute the four partials, writing $c = \cos Q$, $s = \sin Q$ and $\alpha = \sqrt{2P/m\omega}$, $\beta = \sqrt{2Pm\omega}$, so $q = \alpha s$, $p = \beta c$, and $\alpha\beta = 2P$:

$$\frac{\partial q}{\partial Q} = \alpha c, \quad \frac{\partial q}{\partial P} = \frac{s}{\beta}, \quad \frac{\partial p}{\partial Q} = -\beta s, \quad \frac{\partial p}{\partial P} = \frac{c m \omega}{\beta} = \frac{c}{\alpha},$$

using $\partial\alpha/\partial P = \alpha/2P = 1/\beta$ and $\partial\beta/\partial P = \beta/2P = m\omega/\beta$. Then

$$\frac{\partial q}{\partial Q} \frac{\partial p}{\partial P} - \frac{\partial q}{\partial P} \frac{\partial p}{\partial Q} = \alpha c \cdot \frac{c}{\alpha} - \frac{s}{\beta} \cdot (-\beta s) = c^2 + s^2 = 1. \quad \checkmark$$

(This is the Jacobian of the inverse map; since $\det(M^{-1}) = 1/\det M$, the forward Jacobian is 1 too.)

The payoff. Substitute into the oscillator Hamiltonian:

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2 = \frac{2Pm\omega \cos^2 Q}{2m} + \frac{1}{2}m\omega^2 \cdot \frac{2P}{m\omega} \sin^2 Q = \omega P (\cos^2 Q + \sin^2 Q) = \omega P.$$

The Hamiltonian has become a single term with no Q in it. Hamilton's equations are therefore $\dot{Q} = \partial H / \partial P = \omega$ and $\dot{P} = -\partial H / \partial Q = 0$, whose solution is $Q = \omega t + \varphi$ and $P = \text{constant} = E/\omega$. Substituting back gives $q(t) = \sqrt{2E/m\omega^2} \sin(\omega t + \varphi)$. These are the action-angle variables of §4.4's grind box: P is the action $\mathcal{A}/2\pi$, and Q is the angle around the ellipse. The whole solution of the oscillator has been reduced to choosing good coordinates — and the reason we were allowed to choose them is $\det M = 1$.

For $H = p^2/2m + \frac{1}{2}m\omega^2 q^2$, compute the phase-space area enclosed by the orbit of energy E directly from the integral $\oint p \, dq$, and show it equals $2\pi E/\omega$. Confirm it against (1.3.41). Then apply the Bohr–Sommerfeld condition and say precisely what it predicts.

– Solution

The integral. Solving $H = E$ for p gives $p = \pm\sqrt{2mE - m^2\omega^2 q^2}$, with turning points at $q = \pm a$, $a = \sqrt{2E/m\omega^2}$. Going round the loop, the upper branch contributes $\int_{-a}^a$ and the lower contributes the same again with both signs reversed, so

$$\mathcal{A} = \oint p \, dq = 2 \int_{-a}^a \sqrt{2mE - m^2\omega^2 q^2} \, dq = 2m\omega \int_{-a}^a \sqrt{a^2 - q^2} \, dq,$$

using $2mE = m^2\omega^2 a^2$. The remaining integral is the area of a half-disc of radius a , namely $\pi a^2/2$ (substitute $q = a \sin u$ if you want it by hand: $\int_{-\pi/2}^{\pi/2} a^2 \cos^2 u \, du = \pi a^2/2$). Hence

$$\mathcal{A} = 2m\omega \cdot \frac{\pi a^2}{2} = \pi m\omega \frac{2E}{m\omega^2} = \frac{2\pi E}{\omega}.$$

Same answer as the ellipse-area route of Chapter 0.8 §4.4 (π times the product of the semi-axes a and $m\omega a$), as it must be. Numerically, at $m = 1.7$, $\omega = 2.3$, $E = 3$: the semi-axis product route gives 8.19545910 and $2\pi E/\omega$ gives 8.19545910.

The check. $d\mathcal{A}/dE = 2\pi/\omega$, which is the period $T = 2\pi/\omega$. ✓ (1.3.41) holds, and the fact that $\mathcal{A}$ is exactly *linear* in E is the statement that the oscillator's period does not depend on its amplitude — the property that makes it the one system everything is expanded around.

Bohr–Sommerfeld. Impose (1.3.42), $\mathcal{A} = (n + \frac{1}{2})h$:

$$\frac{2\pi E_n}{\omega} = (n + \tfrac{1}{2})h = (n + \tfrac{1}{2})2\pi\hbar \quad \implies \quad \boxed{E_n = (n + \tfrac{1}{2})\hbar\omega.}$$

Three things to notice, all of which Chapter 4.5 will confirm by an exact operator calculation that uses none of this reasoning.

The levels are equally spaced, by $\hbar\omega$. That follows from (1.3.41): equal increments of area correspond to equal increments of energy exactly when the period is energy-independent. A pendulum at large amplitude, whose period grows with energy, has levels that *converge* — and indeed real molecular vibrations, which are anharmonic, have converging levels that eventually run into the dissociation limit. You can read that off the phase portrait without solving anything.

The ground state is not zero. $E_0 = \frac{1}{2}\hbar\omega$, the zero-point energy, enclosing half a unit of h . A classical oscillator can sit at the origin of phase space, a single point of zero area; a quantum one cannot, because the smallest allowed area is $h/2$. Chapter 4.6 says the same thing as $\Delta q \Delta p \geq \hbar/2$ — the uncertainty

principle is the statement that a state occupies a minimum area of phase space, and that minimum is Planck's constant. Every "why can't the electron just fall into the nucleus" question is answered by this picture.

The classical limit is visible. For large n the spacing $\hbar\omega$ is negligible compared with E_n , the rungs of the ladder crowd together, and the discrete set of ellipses is indistinguishable from the continuum.

Numerically, an oscillator with $\omega = 2\pi \text{ s}^{-1}$ carrying 1 J sits at $n \approx 1.5 \times 10^{33}$. This is why nobody noticed.

THE BRICK YOU JUST LAID

You traded n second-order equations for $2n$ first-order ones, by a Legendre transform you now understand geometrically — the dictionary between describing a convex curve by its points and describing it by its tangent lines, the same operation that turns internal energy into free energy. Out of it came the canonical momentum (*not* $m\mathbf{v}$ in general — in a magnetic field it is $m\mathbf{v} + e\mathbf{A}$, and it is that object, not the mechanical momentum, that quantum mechanics turns into an operator), the Hamiltonian (which equals $T + V$ only under two hypotheses you can state), and Hamilton's equations, derived twice.

You then met phase space, in which one point fixes the whole future, trajectories cannot cross, and the pendulum's entire qualitative behaviour is a centre, a saddle, and the separatrix joining them. You proved **Liouville's theorem** in three lines from Chapter 0.7's identity of divergence with the trace of the Jacobian: Hamiltonian flow is exactly incompressible, so information is stirred and never destroyed — and you saw, in the figure, a blob become unrecognisable while its area sat there to within a few parts in ten thousand of integrator error. You saw that the arrow of time is therefore in the coarse-graining, not in the dynamics.

And you built the **Poisson bracket**: bilinear, antisymmetric, Leibniz, Jacobi; $\frac{df}{dt} = \{f, H\} + \frac{\partial f}{\partial t}$ for every observable there is; $\{q^i, p_j\} = \delta_j^i$; and the two-way street of §7, in which a conserved quantity does not merely accompany a symmetry but is the generator of it.

Where this gets spent. The bracket goes to Chapter 4.6, which replaces it by $\frac{1}{i\hbar}[\ , \]$ and changes nothing else — $\{q, p\} = 1$ becomes $[\hat{q}, \hat{p}] = i\hbar$, (1.3.50) becomes the Heisenberg equation — and to Chapter 5.3, which does the same substitution for a field and gets particles. Phase space and Liouville go to Chapter 4.4, where unitarity is the same statement in a Hilbert space, and to statistical mechanics, which cannot define an ensemble without them. The generators of §7 go to Chapter 1.4 (Noether; from the other side), Chapter 4.2 (observables generate unitaries), and Chapter 6.1, where the brackets of generators with each other become a Lie algebra and the whole of Part VI. The canonical momentum goes to Chapter 6.3, where $\mathbf{p} \rightarrow \mathbf{p} - e\mathbf{A}$ is derived from a symmetry principle and named minimal coupling. And Hamilton–Jacobi goes to Chapter 4.6 as the classical limit of the Schrödinger equation — the last stop before the wavefunction.

Chapter 1.4 now closes Part I by proving Noether's theorem in the Lagrangian language, where it applies to fields and to relativity, and where it will be the tool that constructs the conserved currents of every theory in this book.